

SKURAI | 1

López Merino Mauricio $\langle \text{Bra} \mid \text{Ket} \rangle$



$\langle \text{Bra} \mid \text{Ket} \rangle$ ARCHIVE

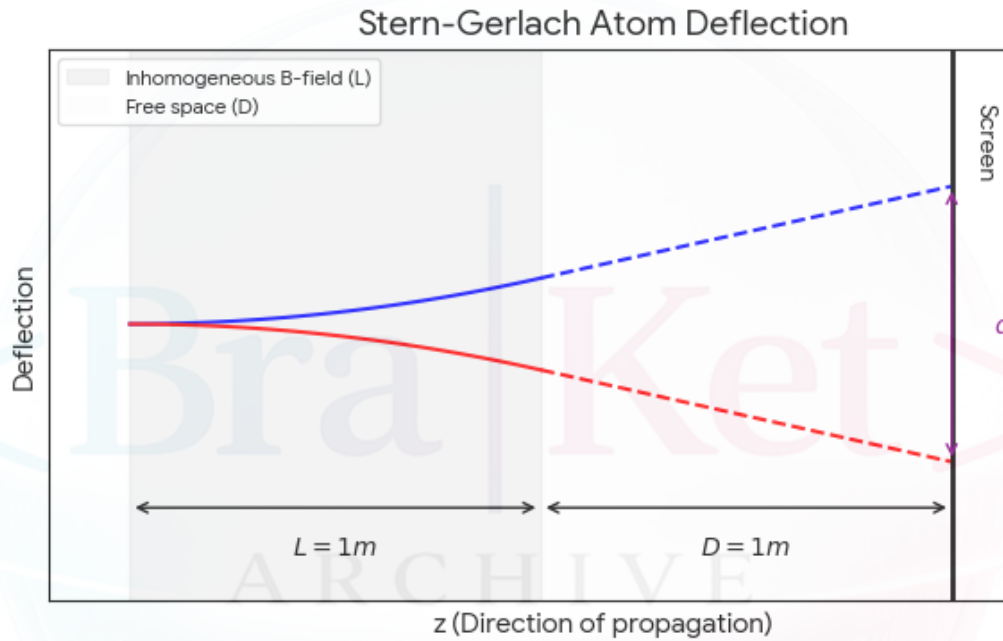
FUNDAMENTAL CONCEPTS
PhD Candidate Preparation - University of Cambridge



Sakurai Problems Chapter one :

- 1.1 A beam of silver atoms is created by heating a vapor in an oven to 1000°C , and selecting atoms with a velocity close to the mean of the thermal distribution. The beam moves through a one-meter long magnetic field with a vertical gradient $10 \frac{T}{m}$, and impinges a screen one meter downstream of the end of the magnet. Assuming the silver atom has spin $\frac{1}{2}$ with a magnetic moment of one Bohr magneton, find the separation distance in millimeters of the two states on the screen.

Solution:



The "Beam" Velocity (A common trap), In Sakurai, when atoms emerge from an oven (effusive beam), we don't just use the standard Maxwell-Boltzmann mean velocity $\langle v \rangle$. Instead, we use the average kinetic energy of the atoms in the beam, which is:

$$\frac{1}{2}mv^2 = 2k_B T,$$

This is higher than the $\frac{3}{2}k_B T$ of the gas inside the oven because faster atoms escape more frequently. Using $v = \sqrt{\frac{4k_B T}{m}}$.

The Equation of Motion can combine the two stages of the trajectory into a single elegant expression. If L is the magnet length and D is the distance to the screen :

$$\Delta z_t = \pm \frac{\mu_B (\nabla B)}{2mv^2} L(L + 2D),$$



Since the problem asks for the separation distance (d) between the two states, you just multiply the absolute displacement by 2:

$$d = \frac{\mu_B (\nabla B)}{mv^2} L(L + 2D),$$

$$d = \frac{\mu_B (\nabla B)}{4k_B T} L(L + 2D),$$

We now:

$$T = 1000^\circ C = 1273.15K,$$

$$\nabla B = 10 \frac{T}{m},$$

$$L = D = 1m,$$

$$\mu_B = 9.274 \times 10^{-24} \frac{J}{T},$$

$$k_B = 1.381 \times 10^{-23} \frac{J}{K},$$

Substitution:

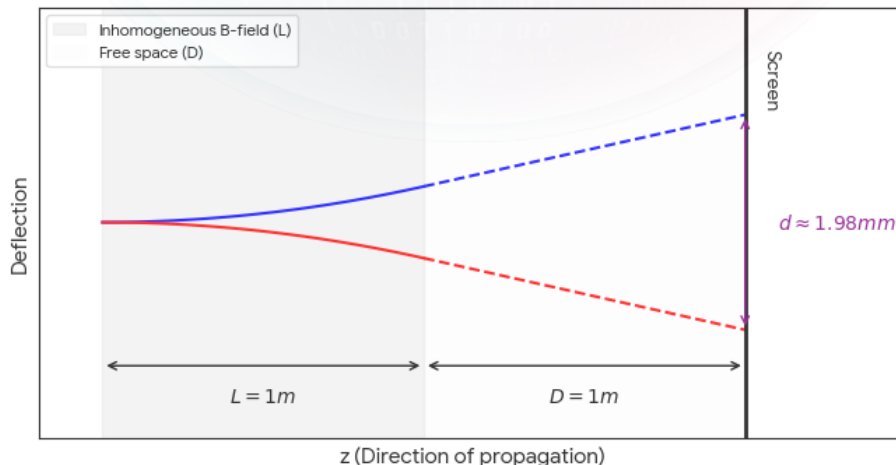
$$d = \frac{\mu_B (\nabla B)}{4k_B T} L(L + 2D) = \frac{\left[9.274 \times 10^{-24} \frac{J}{T} \left(10 \frac{T}{m} \right) \right]}{4 \left[1.381 \times 10^{-23} \frac{J}{K} (1273.15K) \right]} (1m)(1m + 2m),$$

$$d = \frac{\left[9.274 \times 10^{-23} \frac{J}{m} \right]}{4 \left[1.381 \times 10^{-23} J (1273.15) \right]} 3m^2 = \frac{[9.274 \times 10^{-23} J]}{2 \left[1.381 \times 10^{-23} J (1273.15) \right]} m,$$

$$d = \frac{3}{4} \frac{9.274}{1.381(2546.3)} m = (0.75)0.0026373m = 1.978mm,$$

Conclusion:

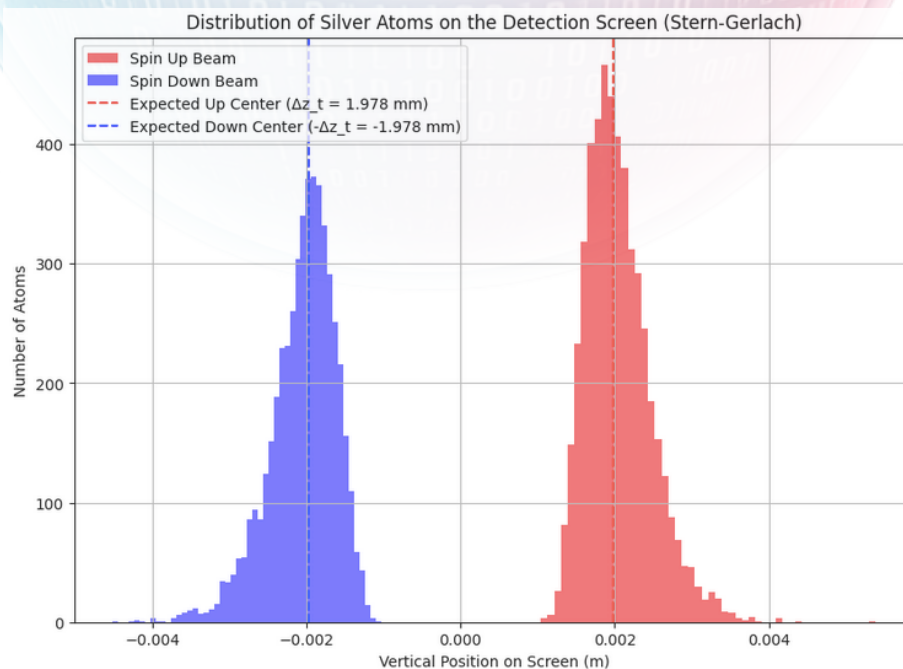
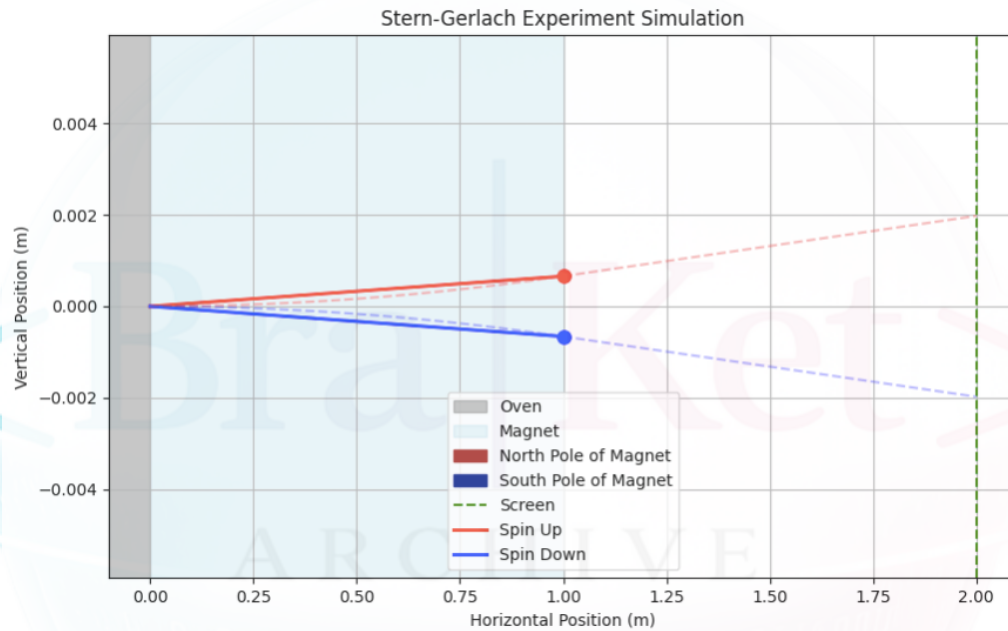
Stern-Gerlach Atom Deflection





The calculated separation of 1.978mm demonstrates the spatial quantization of the silver atom's magnetic moment. The results are consistent with the Stern-Gerlach experiment's prediction, where the atomic beam splits into two discrete components due to the interaction between the intrinsic spin $\left(\frac{1}{2}\right)$ and the external magnetic field gradient.

Montecarlo simulation interactive colab link: https://colab.research.google.com/drive/1-g0_FG_9EgDWzDzLNepHKwqi58fSUYSk?usp=sharing.





1.2 Prove:

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB.$$

Solution:

Remember :

$$[AB, X] = A[B, X] + [A, X]B,$$

Applied :

$$[AB, CD] = A[B, CD] + [A, CD]B,$$

$$[AB, CD] = A(C[B, D] + [B, C]D) + (C[A, D] + [A, C]D)B,$$

$$[AB, CD] = AC[B, D] + A[B, C]D + C[A, D]B + [A, C]DB,$$

$$[AB, CD] = AC([B, D] + 0) + A[B, C]D + C([A, D] + 0)B + [A, C]DB,$$

$$[AB, CD] = AC([B, D] - 2BD + 2BD) + A[B, C]D + C([A, D] - 2AD + 2AD)B + [A, C]DB,$$

Remember :

$$[X, Y] = -[Y, X],$$

$$[X, Y] = \{X, Y\} - 2YX \equiv \{X, Y\} = [X, Y] + 2YX,$$

Applied :

$$[AB, CD] = AC(-[D, B] - 2BD + 2BD) + A[B, C]D + C(-[D, A] - 2AD + 2AD)B + [A, C]DB,$$

$$[AB, CD] = AC(-\{D, B\} + 2BD) + A[B, C]D + C(-\{D, A\} + 2AD)B + [A, C]DB,$$

$$[AB, CD] = -AC\{D, B\} + 2ACBD + A[B, C]D + C[A, D]B + 2CADB + [A, C]DB,$$

$$[AB, CD] = -AC\{D, B\} + A([B, C] + 2CB)D + C\{A, D\}B + (2CA + [A, C])DB,$$

$$[AB, CD] = -AC\{D, B\} + A\{B, C\}D - C\{A, D\}B + \{A, C\}DB,$$

$$\therefore [AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB.$$



1.3 For the spin $\frac{1}{2}$ state $|S_x; +\rangle$, evaluate both sides of the inequality, that is:

$$\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle \geq \frac{1}{4} |\langle [A, B] \rangle|^2,$$

For the operators $A = S_x$ and $B = S_y$, and show that the inequality is satisfied.
Repeat for the operators $A = S_z$ and $B = S_y$.

Solution :

For the operators $A = S_x$ and $B = S_y$, we have :

$$\begin{aligned} \langle (\Delta S_x)^2 \rangle \langle (\Delta S_y)^2 \rangle &\geq \frac{1}{4} |\langle [S_x, S_y] \rangle|^2, \\ \langle (0)^2 \rangle \langle (\Delta S_y)^2 \rangle &\geq \frac{1}{4} |\langle i\hbar S_z \rangle|^2, \\ 0 &\geq \frac{1}{4} |\langle 0 \rangle|^2, \\ 0 &\geq 0, \end{aligned}$$

The inequality is satisfied.

For the operators $A = S_z$ and $B = S_y$, we have:

$$\begin{aligned} \langle (\Delta S_z)^2 \rangle \langle (\Delta S_y)^2 \rangle &\geq \frac{1}{4} |\langle [S_z, S_y] \rangle|^2, \\ \left\langle \frac{\hbar^2}{4} \right\rangle \left\langle \frac{\hbar^2}{4} \right\rangle &\geq \frac{1}{4} |-i\hbar \langle S_x \rangle|^2, \\ \frac{\hbar^4}{16} &\geq \frac{1}{4} \left| -\frac{i\hbar^2}{2} \right|^2, \\ \frac{\hbar^4}{16} &\geq \frac{\hbar^4}{16}, \end{aligned}$$

The inequality is satisfied.

Conclusion:

The equality holds because the state $|S_x; +\rangle$ is a minimum uncertainty state for the non-commuting observables S_y and S_z , which are transverse to the quantization axis.



1.4 Suppose a 2×2 matrix $\vec{X}$ (not necessarily Hermitian, nor unitary) is written as:

$$\vec{X} = a_0 + \vec{\sigma} \cdot \vec{a},$$

Where matrix $\vec{\sigma}$ are :

$$\vec{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \vec{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \vec{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

And a_0 , and $a_{1,2,3}$ are numbers .

(a) How are a_0 and $a_k (k = 1, 2, 3)$ related to $\text{tr}(\vec{X})$ and $\text{tr}(\sigma_k \vec{X})$?

Solution :

$$\text{tr}(\vec{X}) = \text{tr}(a_0 + \sigma_j \cdot a_j) = a_0 \text{tr}(\mathbb{I}) + \text{tr}(\sigma_j) \cdot a_j$$

Remember Consistent with the dimensionality of the $SU(2)$ representation:

$$\text{tr}(\mathbb{I}) = 1 + 1, \text{tr}(\sigma_j) = 1 - 1 = 0,$$

We have :

$$\text{tr}(X) = 2a_0 \Rightarrow \therefore a_0 = \frac{\text{tr}(X)}{2},$$

For other side:

$$\text{tr}(\sigma_k \vec{X}) = \text{tr}(\sigma_k a_0 + \sigma_k \sigma_j a_j) = \text{tr}(\sigma_k a_0 + (\delta_{kj} \mathbb{I} - i\epsilon_{kjl} \sigma_j) a_j),$$

$$\text{tr}(\sigma_k \vec{X}) = \text{tr}(\sigma_k a_0) + \text{tr}((a_k \mathbb{I} - i\epsilon_{kjl} \sigma_j a_j) \cdot) = 0 + 2a_k,$$

$$\text{tr}(\sigma_k \vec{X}) = 2a_k \Rightarrow \therefore a_k = \frac{\text{tr}(\sigma_k \vec{X})}{2},$$

(b) Obtain a_0 and a_k in terms of the matrix elements X_{ij} .

Solution :

$$a_0 = \frac{\text{tr}(X)}{2} = \frac{1}{2} \text{tr} \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} = \frac{1}{2} [X_{11} + X_{22}],$$

$$\therefore a_0 = \frac{1}{2} [X_{11} + X_{22}],$$

For a_k we can see σ_3 is the only one diagonal matrix of σ_k :

$$a_1 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} \right] = \frac{1}{2} \text{tr} \left[\begin{pmatrix} X_{21} & X_{22} \\ X_{11} & X_{12} \end{pmatrix} \right] = \frac{1}{2} [X_{21} + X_{12}],$$

$$a_2 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} \right] = \frac{1}{2} \text{tr} \left[\begin{pmatrix} -iX_{21} & -iX_{22} \\ iX_{11} & iX_{12} \end{pmatrix} \right] = \frac{i}{2} [X_{12} - X_{21}],$$

$$a_3 = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} \right] = \frac{1}{2} \text{tr} \left[\begin{pmatrix} X_{11} & X_{12} \\ -X_{21} & -X_{22} \end{pmatrix} \right] = \frac{1}{2} [X_{11} - X_{22}],$$



1.5 Show that the determinant of a 2×2 matrix $\vec{\sigma} \cdot \vec{a}$ is invariant under :

$$\vec{\sigma} \cdot \vec{a} \rightarrow \vec{\sigma} \cdot \vec{a}' \equiv \exp\left(\frac{i\vec{\sigma} \cdot \hat{n}\phi}{2}\right) \vec{\sigma} \cdot \vec{a} \exp\left(\frac{-i\vec{\sigma} \cdot \hat{n}\phi}{2}\right),$$

Where the matrices $\vec{\sigma}$ are :

$$\vec{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \vec{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \vec{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Find a'_k in terms of a_k when $\hat{n}$ is the positive z-direction and interpret your result.

Solution :

$$U = \exp\left(\frac{i\vec{\sigma} \cdot \hat{n}\phi}{2}\right) \Rightarrow \vec{\sigma} \cdot \vec{a}' = U(\vec{\sigma} \cdot \vec{a})U^\dagger,$$

$$\det(\vec{\sigma} \cdot \vec{a}') = \det(U(\vec{\sigma} \cdot \vec{a})U^\dagger) = \det(U) \det(\vec{\sigma} \cdot \vec{a}) \det(U^\dagger),$$

Since the generator of the transformation $\vec{\sigma} \cdot \hat{n}$ is traceless, the determinant of the exponential operator is unity:

$$\det(U) = e^{\text{tr}\left(\frac{i\vec{\sigma} \cdot \hat{n}\phi}{2}\right)} = e^{\frac{i\phi}{2}\text{tr}(\vec{\sigma} \cdot \hat{n})} = e^{\frac{i\phi}{2}\text{tr}(\sigma_k)\text{tr}(n_k)} = e^{\frac{i\phi}{2}(0)\text{tr}(n_k)} = e^0 = 1,$$

placing U in the $SU(2)$ group, now is trivial to see:

$$\det(\vec{\sigma} \cdot \vec{a}') = \det(U) \det(\vec{\sigma} \cdot \vec{a}) \det(U^\dagger) = (1) \det(\vec{\sigma} \cdot \vec{a}) (1),$$

$$\therefore \det(\vec{\sigma} \cdot \vec{a}') = \det(\vec{\sigma} \cdot \vec{a}),$$

Matrix $\vec{\sigma} \cdot \vec{a}$ is invariant under U .

Now a'_k when $\hat{n} = \hat{z}$:

$$U = \exp\left(\frac{i\vec{\sigma} \cdot \hat{z}\phi}{2}\right) = U = \exp\left(\frac{i\sigma_3\phi}{2}\right) = \cos\left(\frac{\phi}{2}\right)\mathbb{I} + i\sigma_3 \sin\left(\frac{\phi}{2}\right),$$

$$\vec{\sigma} \cdot \vec{a}' = U(\vec{\sigma} \cdot \vec{a})U^\dagger = \left[\cos\left(\frac{\phi}{2}\right)\mathbb{I} + i\sigma_3 \sin\left(\frac{\phi}{2}\right)\right] (\sigma_k a_k) \left[\cos\left(\frac{\phi}{2}\right)\mathbb{I} - i\sigma_3 \sin\left(\frac{\phi}{2}\right)\right],$$

$$\sigma_k a'_k = \left[\cos\left(\frac{\phi}{2}\right) (\sigma_k a_k) + i\sigma_3 (\sigma_k a_k) \sin\left(\frac{\phi}{2}\right)\right] \left[\cos\left(\frac{\phi}{2}\right)\mathbb{I} - i\sigma_3 \sin\left(\frac{\phi}{2}\right)\right],$$

$$\sigma_k a'_k = \left[\cos^2\left(\frac{\phi}{2}\right) (\sigma_k a_k) + i(\sigma_3 (\sigma_k a_k) - (\sigma_k a_k) \sigma_3) \sin\left(\frac{\phi}{2}\right) \cos\left(\frac{\phi}{2}\right) + \sigma_3 (\sigma_k a_k) \sigma_3 \sin^2\left(\frac{\phi}{2}\right)\right],$$



$$\sigma_k a'_k = \left[\cos^2 \left(\frac{\phi}{2} \right) (\sigma_k a_k) + i[\sigma_3, \sigma_k a_k] \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) + a_k (\sigma_3 \sigma_k \sigma_3) \sin^2 \left(\frac{\phi}{2} \right) \right],$$

Remember:

$$\{\sigma_i | \sigma_j\} = \sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \mathbb{I} \Rightarrow \sigma_i \sigma_j = 2\delta_{ij} \mathbb{I} - \sigma_j \sigma_i \Rightarrow \sigma_i \sigma_j \sigma_i = (2\delta_{ij} \mathbb{I} - \sigma_j \sigma_i) \sigma_i,$$

$$\therefore \sigma_i \sigma_j \sigma_i = 2\delta_{ij} \sigma_i - \sigma_j;$$

$$\sigma_k a'_k = \left[\cos^2 \left(\frac{\phi}{2} \right) (\sigma_k a_k) + i[2i\epsilon_{3kl} \sigma_l a_k] \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) + a_k (2\delta_{3k} \sigma_3 - \sigma_k) \sin^2 \left(\frac{\phi}{2} \right) \right],$$

$$\sigma_k a'_k = \left[\cos^2 \left(\frac{\phi}{2} \right) (\sigma_k a_k) - 2[\epsilon_{3kl} \sigma_l a_k] \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) + a_k (2\delta_{3k} \sigma_3 - \sigma_k) \sin^2 \left(\frac{\phi}{2} \right) \right],$$

Now a'_k :

$$\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix} = \cos^2 \left(\frac{\phi}{2} \right) \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} - 2 \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) \begin{pmatrix} a_2 \\ -a_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -a_1 \\ -a_2 \\ a_3 \end{pmatrix} \sin^2 \left(\frac{\phi}{2} \right),$$

$$\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix} = \left(\cos^2 \left(\frac{\phi}{2} \right) - \sin^2 \left(\frac{\phi}{2} \right) \right) \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix} - 2 \sin \left(\frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) \begin{pmatrix} a_2 \\ -a_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ a_3 \end{pmatrix},$$

Applying standard double-angle relations, we obtain the transformed vector components in terms of the physical rotation angle ϕ :

$$\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix} = \cos(\phi) \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix} - \sin(\phi) \begin{pmatrix} a_2 \\ -a_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ a_3 \end{pmatrix},$$

$$\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix} = \begin{pmatrix} \cos(\phi) & -\sin(\phi) & 0 \\ \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix},$$

This result confirms that while the spin-1/2 state (spinor) transforms under $SU(2)$ with an angle $\frac{\phi}{2}$, the corresponding expectation values of the spin operators (vectors) transform under the $SO(3)$ rotation group with the full angle ϕ .



1.6 Using the rules of bra-ket algebra, prove or evaluate the following:

- a. $\text{tr}(XY) = \text{tr}(YX)$, where X and Y are operators;

Solution:

$$\begin{aligned}\text{tr}(XY) &= \sum_{a'} \langle a' | XY | a' \rangle = \sum_{a'} \sum_{a''} \langle a' | X | a'' \rangle \langle a'' | Y | a' \rangle, \\ \text{tr}(XY) &= \sum_{a''} \sum_{a'} \langle a'' | Y | a' \rangle \langle a' | X | a'' \rangle = \sum_{a''} \langle a'' | YX | a'' \rangle, \\ &\therefore \text{tr}(XY) = \text{tr}(YX).\end{aligned}$$

- b. $(XY)^\dagger = Y^\dagger X^\dagger$, where X and Y are operators;

Solution:

Consider the inner product formed by the action of the product operator XY on an arbitrary ket $|\psi\rangle$:

$$|\psi\rangle = XY|\phi\rangle,$$

By the fundamental definition of the adjoint operator, the corresponding bra is:

$$\langle\psi| = \langle\phi|(XY)^\dagger,$$

On the other hand, we can evaluate $\langle\psi|$ by applying the operators sequentially.

Let $\langle\alpha| = \langle\phi|Y^\dagger$:

$$\langle\psi| = \langle\alpha|X^\dagger,$$

$$\langle\psi| = (\langle\phi|Y^\dagger)X^\dagger,$$

$$\therefore (XY)^\dagger = Y^\dagger X^\dagger.$$

- c. $\exp(if(A)) = ?$ In ket-bra form, where A is a Hermitian operator whose eigenvalues are known;

Solution :

Since A a Hermitian operator, its eigenstates form a complete orthonormal basis.

We can express the operator A through its spectral decomposition:

$$A = \sum_{a'} a' |a'\rangle \langle a'|,$$

For any function $g(A)$ that can be expanded in a Taylor series, the operator acts on its eigenstates such that $g(A)|a'\rangle = g(a')|a'\rangle$. Therefore, the ket-bra representation of the function is:

$$g(A) = \sum_{a'} g(a') |a'\rangle \langle a'|,$$



Substituting $g(A) = \exp(if(A))$, we obtain:

$$\exp(if(A)) = \sum_{a'} \exp(if(a')) |a'\rangle\langle a'|.$$

d. $\sum_{a'} \psi_{a'}^*(\mathbf{x}') \psi_{a'}(\mathbf{x}'')$, where $\psi_{a'}(\mathbf{x}') = \langle \mathbf{x}' | a' \rangle$.

Solution:

Substitute the wave function definition into the summation:

$$\sum_{a'} \psi_{a'}^*(\mathbf{x}') \psi_{a'}(\mathbf{x}'') = \sum_{a'} \langle a' | \mathbf{x}' \rangle \langle \mathbf{x}'' | a' \rangle,$$

Since these are inner products (complex numbers), we can rearrange their order:

$$\sum_{a'} \langle \mathbf{x}'' | a' \rangle \langle a' | \mathbf{x}' \rangle,$$

By applying the closure relation (completeness) for the eigenstates the expression collapses to:

$$\langle \mathbf{x}'' | \left(\sum_{a'} |a'\rangle\langle a'| \right) | \mathbf{x}' \rangle = \langle \mathbf{x}'' | \mathbf{x}' \rangle,$$

By the definition of the position basis orthonormality in a continuous representation, this inner product is the Dirac delta function:

$$\therefore \sum_{a'} \psi_{a'}^*(\mathbf{x}') \psi_{a'}(\mathbf{x}'') = \delta(\mathbf{x}'' - \mathbf{x}').$$



1.7 a. Consider two kets $|\alpha\rangle$ and $|\beta\rangle$. Suppose $\langle a'|\alpha\rangle, \langle a''|\alpha\rangle, \dots$ and $\langle a'|\beta\rangle, \langle a''|\beta\rangle, \dots$ are all known, where $|a'\rangle, |a''\rangle, \dots$ form a complete set of base kets. Find the matrix representation of the operator $|\alpha\rangle\langle\beta|$ in that basis.

Solution:

By definition, the elements of a matrix representing an operator X in the basis $\{|a'\rangle\}$ are given by $X_{ij} = \langle a^{(i)}|X|a^{(j)}\rangle$.

Substituting the operator $X = |\alpha\rangle\langle\beta|$, we have:

$$X_{ij} = \langle a^{(i)}|(|\alpha\rangle\langle\beta|)|a^{(j)}\rangle,$$

Since the inner products are complex numbers (scalars), we can regroup the expression:

$$X_{ij} = \langle a^{(i)}|\alpha\rangle\langle\beta|a^{(j)}\rangle,$$

Recognizing the property of the dual correspondence $\langle\beta|a^{(j)}\rangle = \langle a^{(j)}|\beta\rangle^*$, the matrix elements are:

$$X_{ij} = \langle a^{(i)}|\alpha\rangle\langle a^{(j)}|\beta\rangle^*,$$

This shows that the matrix is formed by the outer product of the column vector representing $|\alpha\rangle$ and the complex conjugate of the row vector representing $|\beta\rangle$.

b. We now consider a spin 1/2. System and let $|\alpha\rangle$ and $|\beta\rangle$ be $|S_z; +\rangle$ and $|S_x; +\rangle$, respectively. Write down explicitly the square matrix that corresponds to $|\alpha\rangle\langle\beta|$ in the usual (S_z diagonal) basis.

Solution:

We are working in the usual S_z basis: $\{|+\rangle, |-\rangle\}$.

The given kets are:

$$|\alpha\rangle = |S_z; +\rangle = |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$|\beta\rangle = |S_x; +\rangle = \frac{1}{\sqrt{2}}[|+\rangle + |-\rangle] = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

To find the matrix $M = |\alpha\rangle\langle\beta|$, we first determine the bra $\langle\beta|$:

$$\langle\beta| = \frac{1}{\sqrt{2}}[\langle+| + \langle-|],$$

Now, we perform the outer product:

$$M = |\alpha\rangle\langle\beta| = |+\rangle \left[\frac{1}{\sqrt{2}}[\langle+| + \langle-|] \right] = \frac{1}{\sqrt{2}}[|+\rangle\langle+| + |+\rangle\langle-|],$$



In the $\{|+\rangle, |-\rangle\}$ basis (the S_z representation), the operator is expressed as:

$$\therefore M \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

This is a not Hermitian matrix ($M \neq M^\dagger$). In the context of quantum mechanics, this operator does not represent a physical observable unless $|\alpha\rangle = |\beta\rangle$.

1.8 Suppose $|i\rangle$ and $|j\rangle$ are eigenkets of some Hermitian operator A . Under what condition can we conclude that $|i\rangle + |j\rangle$ is also an eigenket of A ? Justify your Answer.

Solution :

Let $|i\rangle$ and $|j\rangle$ be eigenkets of the Hermitian operator A with eigenvalues a_i and a_j respectively:

$$A|i\rangle = a_i|i\rangle \text{ and } A|j\rangle = a_j|j\rangle,$$

Consider the linear combination $|\Psi\rangle = c_i|i\rangle + c_j|j\rangle$, where $c_i, c_j \in \mathbb{C}$.

For $|\Psi\rangle$ to be an eigenket of A , there must exist an eigenvalue λ such that:

$$A(c_i|i\rangle + c_j|j\rangle) = \lambda(c_i|i\rangle + c_j|j\rangle),$$

Applying the linearity of the operator A :

$$c_i A|i\rangle + c_j A|j\rangle = c_i \lambda|i\rangle + c_j \lambda|j\rangle,$$

$$c_i a_i|i\rangle + c_j a_j|j\rangle = c_i \lambda|i\rangle + c_j \lambda|j\rangle,$$

$$a_i|i\rangle + a_j|j\rangle = \lambda|i\rangle + \lambda|j\rangle,$$

$$(a_i - \lambda)|i\rangle + (a_j - \lambda)|j\rangle = 0,$$

Assuming $|i\rangle$ and $|j\rangle$ are linearly independent (which is generally the case for distinct eigenkets), the only way for this linear combination to vanish is if each coefficient is zero:

$$\begin{cases} a_i - \lambda = 0 \\ a_j - \lambda = 0 \end{cases} \Rightarrow \therefore a_i = a_j = \lambda.$$

Therefore, the condition is that $|i\rangle$ and $|j\rangle$ must be degenerate, meaning they belong to the same eigensubspace associated with the eigenvalue λ . Under this condition, any vector in the $\text{span}\{|i\rangle, |j\rangle\}$ is also an eigenket of A .



1.9 Consider a ket space spanned by the eigenkets $\{|a'\rangle\}$ of a Hermitian operator A . There is no degeneracy.

a. Prove that :

$$\prod_{a'} (A - a'),$$

Is the null operator.

Solution :

Since A is a Hermitian operator with no degeneracy, its eigenkets $\{|a'\rangle\}$ form a complete orthonormal basis. Any state $|\Psi\rangle$ can be expanded as:

$$|\Psi\rangle = \sum_{a_i} c_i |a_i\rangle,$$

Let $\hat{\mathcal{P}} = \prod_{a'} (A - a')$. Since all factors are functions of the same operator A , they commute with one another. To find the effect on a specific basis ket $|a_k\rangle$, we rearrange the product:

$$\hat{\mathcal{P}}|a_k\rangle = \left[\prod_{a' \neq a_k} (A - a') \right] (A - a_k)|a_k\rangle,$$

Annihilation: By the definition of an eigenvalue, $(A - a_k)|a_k\rangle = 0$. Thus:

$$\hat{\mathcal{P}}|a_k\rangle = \left[\prod_{a' \neq a_k} (A - a') \right] 0 = 0,$$

Since $\hat{\mathcal{P}}$ annihilates every basis vector, it must annihilate any linear combination $|\Psi\rangle$. Therefore, $\hat{\mathcal{P}}$ is the null operator. This is the operator-theoretic version of the Cayley-Hamilton theorem.

b. What is the significance of :

$$\prod_{a'' \neq a'} \frac{(A - a'')}{(a' - a'')},$$

Solution:

This is the Projection Operator $|a'\rangle\langle a'|$. Its significance is understood by observing its action on the basis:

- Case $k = a'$: Every term in the product becomes $\frac{(a' - a'')}{(a' - a'')} = 1$. Thus, $P_{a'}|a'\rangle$.
- Case $k \neq a'$: There exists one term in the product where $a'' = a_k$. For this specific term, the numerator becomes $(A - a_k)|a_k\rangle = 0$, causing the entire product to vanish. Thus, $P_{a'}|a_k\rangle = 0$.



$P_{a'}$ acts as a "filter" that selects only the component corresponding to the eigenvalue a' . This construction is the operator equivalent of the Lagrange Interpolating Polynomial.

- c. Illustrate (a) and (b) using A set equal to S_z of a spin $\frac{1}{2}$ system.

Solution:

For a spin $\frac{1}{2}$ system, the eigenvalues are $a' \in \left\{\frac{\hbar}{2}, -\frac{\hbar}{2}\right\}$ with eigenkets $|+\rangle$ and $|-\rangle$.

Part (a) illustration:

$$\hat{\mathcal{P}} = \left(S_z - \frac{\hbar}{2}\right)\left(S_z + \frac{\hbar}{2}\right),$$

Applying this to the basis kets:

$$\hat{\mathcal{P}}|+\rangle = \left(S_z + \frac{\hbar}{2}\right)\left(S_z - \frac{\hbar}{2}\right)|+\rangle = 0,$$

$$\hat{\mathcal{P}}|-\rangle = \left(S_z - \frac{\hbar}{2}\right)\left(S_z + \frac{\hbar}{2}\right)|-\rangle = 0,$$

As it annihilates the entire basis, $\hat{\mathcal{P}} = 0$.

Part (b) Illustration (Projector for $|+\rangle$):

$$P_+ = \frac{S_z - \left(-\frac{\hbar}{2}\mathbb{I}\right)}{\frac{\hbar}{2} - \left(-\frac{\hbar}{2}\right)} = \frac{S_z + \frac{\hbar}{2}\mathbb{I}}{\hbar},$$

Using the spectral decomposition $S_z = \frac{\hbar}{2}|+\rangle\langle+| - \frac{\hbar}{2}|-\rangle\langle-|$ and $\mathbb{I} = |+\rangle\langle+| + |-\rangle\langle-|$:

$$P_+ = \frac{\left(\frac{\hbar}{2} + \frac{\hbar}{2}\right)|+\rangle\langle+| + \left(-\frac{\hbar}{2} + \frac{\hbar}{2}\right)|-\rangle\langle-|}{\hbar} = \frac{\hbar|+\rangle\langle+|}{\hbar} = |+\rangle\langle+|.$$

1.10 Using the orthonormality of $|+\rangle$ and $|-\rangle$, prove:

$$[S_i, S_j] = i\varepsilon_{ijk}\hbar S_k, \quad \{S_i, S_j\} = \left(\frac{\hbar^2}{2}\right)\delta_{ij},$$

Where:

$$S_x = \frac{\hbar}{2}(|+\rangle\langle-| + |-\rangle\langle+|), \quad S_y = \frac{i\hbar}{2}(-|+\rangle\langle-| + |-\rangle\langle+|),$$

$$S_z = \frac{\hbar}{2}(|+\rangle\langle+| - |-\rangle\langle-|).$$

Solution:

Proof of the Commutation Relation $[S_i, S_j] = i\varepsilon_{ijk}\hbar S_k$:

We demonstrate the rigor by computing $[S_x, S_y]$. Due to the rotational symmetry of the space, this result generalizes to all cyclic permutations.

Step 1: Compute the product $S_x S_y$:

$$S_x S_y = \left(\frac{\hbar}{2}\right)\left(\frac{i\hbar}{2}\right)[|+\rangle\langle-| + |-\rangle\langle+|][|-\rangle\langle+| + |+\rangle\langle-|],$$



Using orthonormality ($\langle -|+\rangle = 0$ and $\langle +|+\rangle = \langle -|-\rangle = 1$), we expand the terms:

1. $[|+\rangle\langle -|][-|+\rangle\langle -|] = -|+\rangle\langle -|+\rangle\langle -| = 0$,
2. $[|+\rangle\langle -|][-|-\rangle\langle +|] = |+\rangle\langle -|-\rangle\langle +| = |+\rangle\langle +|$,
3. $[|- \rangle\langle +|][-|+\rangle\langle -|] = -|- \rangle\langle +|+\rangle\langle -| = -|- \rangle\langle -|$,
4. $[|- \rangle\langle +|][-|-\rangle\langle +|] = |- \rangle\langle +|-\rangle\langle +| = 0$,

Thus, $S_x S_y = \frac{i\hbar^2}{4} [|+\rangle\langle +| - |- \rangle\langle -|]$,

Step 2: Compute the product $S_y S_x$:

Following the same logic:

$$S_y S_x = \frac{i\hbar^2}{4} [-|+\rangle\langle -| + |- \rangle\langle +|][|+\rangle\langle -| + |- \rangle\langle +|],$$

$$S_y S_x = \frac{i\hbar^2}{4} [-|+\rangle\langle +| + |- \rangle\langle -|],$$

Step 3: Evaluate the Commutator:

$$[S_x, S_y] = S_x S_y - S_y S_x = \frac{i\hbar^2}{2} [|+\rangle\langle +| - |- \rangle\langle -|] = i\hbar \left[\frac{\hbar}{2} |+\rangle\langle +| - |- \rangle\langle -| \right],$$

$$\therefore [S_x, S_y] = i\varepsilon_{xyz} \hbar S_z.$$

Proof of the Anticommutation Relation $\{S_i, S_j\} = \left(\frac{\hbar^2}{2}\right) \delta_{ij}$:

Case 1 Diagonal terms ($i = j$):

We calculate S_z^2 :

$$S_z^2 = \frac{\hbar^2}{4} [|+\rangle\langle +| - |- \rangle\langle -|]^2,$$

Due to orthogonality, the cross-terms vanish:

$$S_z^2 = \frac{\hbar^2}{4} [|+\rangle\langle +| + |- \rangle\langle -|] = \frac{\hbar^2}{4} \mathbb{I},$$

The anticommutator is $\{S_z, S_z\} = 2S_z^2 = \frac{\hbar^2}{2} \mathbb{I}$, satisfying the identity for $\delta_{zz} = \mathbb{I}$.

Case 2 Off-diagonal terms ($i \neq j$):

Using our results from section 2:

$$S_x S_y = \frac{i\hbar^2}{4} [|+\rangle\langle +| - |- \rangle\langle -|],$$

$$S_y S_x = -\frac{i\hbar^2}{4} [|+\rangle\langle +| - |- \rangle\langle -|],$$

The sum yields:

$$\{S_x, S_y\} = S_x S_y + S_y S_x = 0.$$

This satisfies the identity as $\delta_{xy} = 0$.

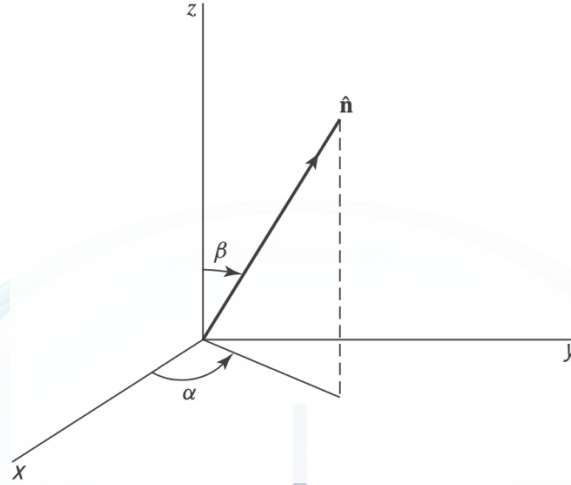
The operators satisfy the Lie Algebra of $SU(2)$ through their commutation relations and generate a representation of the Clifford Algebra $\mathcal{Cl}_{3,0}(\mathbb{R})$ through their anticommutation relations. This duality ensures that while the operators generate rotations, they also represent the physical constraint of the spin-1/2 state space.



1.11 Construct $|\vec{S} \cdot \hat{n}; +\rangle$ such that

$$\vec{S} \cdot \hat{n} |\vec{S} \cdot \hat{n}; +\rangle = \left(\frac{\hbar}{2}\right) |\vec{S} \cdot \hat{n}; +\rangle,$$

Where $\hat{n}$ is characterized by the angles shown in the figure. Express your answer as a linear combination of $|+\rangle$ and $|-\rangle$.



Solution:

To solve this as a straightforward eigenvalue problem, we first project the spin operator $\vec{S}$ onto the unit vector $\hat{n}$ and then solve the characteristic equation without invoking the rotation group $SO(3)$ or its double cover $SU(2)$.

The unit vector $\hat{n}$ in spherical coordinates is given by:

$$\hat{n} = \begin{pmatrix} \sin(\beta) \cos(\alpha) \\ \sin(\beta) \sin(\alpha) \\ \cos(\beta) \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix},$$

The operator $\vec{S} \cdot \hat{n}$ is constructed using the components S_x, S_y, S_z :

$$\vec{S} \cdot \hat{n} = \begin{pmatrix} \sin(\beta) \cos(\alpha) \\ \sin(\beta) \sin(\alpha) \\ \cos(\beta) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix},$$

Instead of a matrix, we express the operator $\vec{S} \cdot \hat{n}$ using the ladder operators $S_{\pm} = S_x \pm iS_y$:

$$\vec{S} \cdot \hat{n} = S_z \cos(\beta) + \frac{1}{2} \sin(\beta) (S_+ e^{-i\alpha} + S_- e^{i\alpha}),$$

Let $|\hat{n}; +\rangle = a |+\rangle + b |-\rangle$. The eigenvalue equation is:

$$\left[S_z \cos(\beta) + \frac{1}{2} \sin(\beta) (S_+ e^{-i\alpha} + S_- e^{i\alpha}) \right] (a |+\rangle + b |-\rangle) = \frac{\hbar}{2} (a |+\rangle + b |-\rangle),$$



Applying the operators:

$$\begin{aligned} S_z|+\rangle &= \frac{\hbar}{2}|+\rangle, & S_z|-\rangle &= -\frac{\hbar}{2}|-\rangle, \\ S_+|-\rangle &= \hbar|+\rangle, & S_+|+\rangle &= 0, \\ S_-|+\rangle &= \hbar|-\rangle, & S_-|-\rangle &= 0, \end{aligned}$$

Substitute these into the equation and project onto $\langle +|$ and $\langle -|$:

Projection onto $\langle +|$:

$$\begin{aligned} a \frac{\hbar}{2} \cos(\beta) + b \frac{\hbar}{2} \sin(\beta) e^{-i\alpha} &= \frac{\hbar}{2} a, \\ \therefore \frac{b}{a} &= \frac{1 - \cos(\beta)}{\sin(\beta)} e^{i\alpha}, \end{aligned}$$

Projection onto $\langle -|$:

$$\begin{aligned} -b \frac{\hbar}{2} \cos(\beta) + a \frac{\hbar}{2} \sin(\beta) e^{i\alpha} &= \frac{\hbar}{2} b, \\ \therefore \frac{b}{a} &= \frac{\sin(\beta)}{1 + \cos(\beta)} e^{i\alpha}, \end{aligned}$$

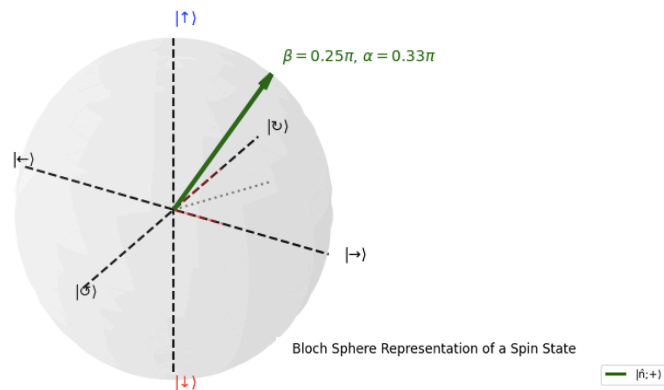
Using the identity $\tan\left(\frac{\beta}{2}\right) = \frac{1 - \cos(\beta)}{\sin(\beta)} = \frac{\sin(\beta)}{1 + \cos(\beta)}$, both projections yield:

$$\frac{b}{a} = \tan\left(\frac{\beta}{2}\right) e^{i\alpha},$$

By setting $a = \cos\left(\frac{\beta}{2}\right)$ to satisfy $\langle \Psi | \Psi \rangle = 1$:

$$\therefore |\hat{n}; +\rangle = \cos\left(\frac{\beta}{2}\right) |+\rangle + \sin\left(\frac{\beta}{2}\right) e^{i\alpha} |-\rangle.$$

Notice that even though the physical direction $\hat{n}$ is periodic in 2π for β , the state vector only returns to its original value after a 4π rotation (due to the $\beta/2$ factor). This is the fundamental definition of a spinor and illustrates why $SU(2)$ is the universal cover of $SO(3)$.





1.12 The Hamiltonian operator for a two-state system is given by:

$$H = a(|1\rangle\langle 1| - |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|),$$

Where a is a number with the dimension of energy. Find the energy eigenvalues and the corresponding energy eigenkets (as linear combinations of $|1\rangle$ and $|2\rangle$).

Solution:

We define the following operators based on the $\{|1\rangle, |2\rangle\}$ basis:

Inversion Operator: $\sigma_z = |1\rangle\langle 1| - |2\rangle\langle 2|$,

Transition Operators: $\sigma_x = |1\rangle\langle 2| + |2\rangle\langle 1|$,

Thus, the Hamiltonian is compactly expressed as:

$$H = a(\sigma_z + \sigma_x),$$

The eigenvalues of any operator A satisfy the relation $\lambda^2 \in \text{eigenvalues } A^2$:

$$H^2 = a^2(\sigma_z + \sigma_x)^2 = a^2[\sigma_z^2 + \sigma_x^2 + \{\sigma_z, \sigma_x\}],$$

From the fundamental properties of two-level systems (Clifford Algebra):

$$\sigma_z^2 = [|1\rangle\langle 1| - |2\rangle\langle 2|]^2 = |1\rangle\langle 1| + |2\rangle\langle 2| = \mathbb{I},$$

$$\sigma_x^2 = [|1\rangle\langle 2| + |2\rangle\langle 1|]^2 = |1\rangle\langle 1| + |2\rangle\langle 2| = \mathbb{I},$$

Anticommutator $\{\sigma_z, \sigma_x\} = 0$ (as proven in previous exercises).

$$H^2 = 2a^2\mathbb{I},$$

The eigenvalues E must satisfy $E^2 = 2a^2$, leading directly to:

$$E = \pm a\sqrt{2},$$

Using the Spectral Projection method, the eigenkets corresponding to $E_{\pm} = \pm a\sqrt{2}$ can be generated by acting the projection operator $P_{\pm} = \frac{1}{2}\left(1 \pm \frac{H}{E_{\pm}}\right)$ on an arbitrary state, say $|1\rangle$:

$$|E_{\pm}\rangle \propto \left(1 \pm \frac{a(\sigma_z + \sigma_x)}{a\sqrt{2}}\right)|1\rangle,$$

$$|E_{\pm}\rangle \propto |1\rangle \pm \frac{1}{\sqrt{2}}[\sigma_z|1\rangle + \sigma_x|1\rangle],$$

Using $\sigma_z|1\rangle = |1\rangle$, and $\sigma_x|1\rangle = |2\rangle$:

$$|E_{\pm}\rangle \propto |1\rangle \pm \frac{1}{\sqrt{2}}[|1\rangle + |2\rangle],$$

$$\therefore |E_{\pm}\rangle \propto \left(1 \pm \frac{1}{\sqrt{2}}\right)|1\rangle \pm \frac{1}{\sqrt{2}}|2\rangle,$$



To determine the normalization constant $N_{\pm}$ for the states $|E_{\pm}\rangle$, we impose the condition $\langle E_{\pm}|E_{\pm}\rangle = 1$.

Given our previous result:

$$|E_{\pm}\rangle = N_{\pm} \left[\left(1 \pm \frac{1}{\sqrt{2}}\right) |1\rangle \pm \frac{1}{\sqrt{2}} |2\rangle \right],$$

The inner product is:

$$\begin{aligned} \langle E_{\pm}|E_{\pm}\rangle &= N_{\pm}^2 \left[\left(\frac{2 \pm \sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 \right], \\ \langle E_{\pm}|E_{\pm}\rangle &= N_{\pm}^2 [2 \pm \sqrt{2}], \end{aligned}$$

Setting this to unity, we find:

$$N_{\pm} = \frac{1}{\sqrt{2 \pm \sqrt{2}}}$$

The complete energy eigenkets are:

$$\therefore |E_{\pm}\rangle = \frac{1}{\sqrt{2 \pm \sqrt{2}}} \left[\frac{2 \pm \sqrt{2}}{2} |1\rangle \pm \frac{\sqrt{2}}{2} |2\rangle \right],$$

These constants might look messy, but they are exactly what you get when you calculate the half-angle of the mixing angle in a Hamiltonian that is a sum of σ_x and σ_z . In this case, the "field" points at 45° , and the spinor follows half that angle.

1.13 A two-state system is characterized by the Hamiltonian:

$$H = H_{11}|1\rangle\langle 1| + H_{22}|2\rangle\langle 2| + H_{12}[|1\rangle\langle 2| + |2\rangle\langle 1|],$$

Where H_{11} , H_{22} and H_{12} are real numbers with the dimension of energy, and $|1\rangle$ and $|2\rangle$ are eigenkets of some observable ($\neq H$). Find the energy eigenkets and corresponding energy eigenvalues. Make sure that your answer makes good sense for $H_{12} = 0$.

Solution :

Any 2×2 Hermitian matrix can be written as a linear combination of the identity $\mathbb{I}$ and the Pauli operators $\vec{\sigma}$. Let's define:

$$\text{Average Energy: } \bar{H} = \frac{1}{2} [H_{11} + H_{22}],$$

$$\text{Energy Difference: } \Delta = \frac{1}{2} [H_{11} - H_{22}],$$

We can rewrite the Hamiltonian as:

$$H = \bar{H} [|1\rangle\langle 1| + |2\rangle\langle 2|] + \Delta [|1\rangle\langle 1| - |2\rangle\langle 2|] + H_{12} [|1\rangle\langle 2| + |2\rangle\langle 1|],$$



$$H = \bar{H}\mathbb{I} + \Delta\sigma_z + H_{12}\sigma_x,$$

The eigenvalues of an operator of the form $A = c_0\mathbb{I} + \vec{c} \cdot \vec{\sigma}$ are simply $E = c_0 \pm |\vec{c}|$. In our case, $c_0 = \bar{H}$ and $\vec{c} = (H_{12}, 0, \Delta)$. Therefore :

$$E_{\pm} = \bar{H} \pm \sqrt{\Delta^2 + H_{12}^2},$$

Substituting the original variables:

$$E_{\pm} = \frac{1}{2}[H_{11} + H_{22}] \pm \sqrt{\frac{1}{4}[H_{11} - H_{22}]^2 + H_{12}^2},$$

If $H_{12} = 0$, then $E_{\pm} = \frac{1}{2}[H_{11} + H_{22}] \pm \frac{1}{2}|H_{11} - H_{22}|$.

This yields $E_+ = H_{11}$ and $E_- = H_{22}$ (assuming $H_{11} > H_{22}$), which makes perfect physical sense as the system becomes uncoupled.

An energy eigenket $|E\rangle$ satisfies $(H - E\mathbb{I})|E\rangle = 0$. Let $|E\rangle = c_1|1\rangle + c_2|2\rangle$. Projecting this equation onto the bra $\langle 1|$, we obtain :

$$(H_{11} - E)c_1 + H_{12}c_2 = 0 \Rightarrow \frac{c_2}{c_1} = \frac{E - H_{11}}{H_{12}},$$

Instead of substituting the long expression for E into the coefficients, we express the kets in terms of the energy shift relative to the site energy. Defining the mixing ratio $\eta_{\pm} = \frac{E_{\pm} - H_{11}}{H_{12}}$, the normalized eigenkets are:

$$\therefore |E_{\pm}\rangle = \frac{1}{\sqrt{1 + |\eta_{\pm}|^2}}[|1\rangle + \eta_{\pm}|2\rangle],$$

Avoided Crossing: The presence of the off-diagonal coupling H_{12} prevents energy level degeneracy. Even at the resonance point ($H_{11} = H_{22}$), a minimum energy gap of $2|H_{12}|$ is maintained, illustrating the von Neumann-Wigner theorem.

Universal Mixing: The mixing ratio $\eta_{\pm}$ quantifies the transition from localized states (when $|H_{12}| \ll |H_{11} - H_{22}|$) to maximal delocalization (when $H_{11} = H_{22}$), where the eigenkets become symmetric and antisymmetric Bell states: $|E_{\pm}\rangle = \frac{1}{\sqrt{2}}[|1\rangle \pm |2\rangle]$.



1.14 A spin $\frac{1}{2}$ system is known to be in an eigenstate of $\vec{S} \cdot \hat{n}$ with eigenvalue $\hbar/2$, where $\hat{n}$ is a unit vector lying in the xz -plane that makes an angle γ with the positive z -axis.

a. Suppose S_x is measured. What is the probability of getting $+\hbar/2$?

b. Evaluate the dispersion in S_x , that is,

$$\langle (S_x - \langle S_x \rangle)^2 \rangle.$$

(For your own peace of mind check your answers for the special cases $\gamma = 0, \pi/2$, and π .)

Solution:

The unit vector is defined as $\hat{n} = \sin(\gamma) \hat{x} + \cos(\gamma) \hat{z}$. The eigenstate of $\vec{S} \cdot \hat{n}$ with eigenvalue $\hbar/2$ is:

$$|\hat{n}; +\rangle = \frac{1}{\sqrt{2}} \left[\cos\left(\frac{\gamma}{2}\right) |+\rangle + \sin\left(\frac{\gamma}{2}\right) |-\rangle \right],$$

We want the probability of obtaining $+\hbar/2$ when measuring S_x . The corresponding eigenstate for S_x is :

$$|S_x; +\rangle = \frac{1}{\sqrt{2}} [|+\rangle + |-\rangle],$$

The probability is the square of the transition amplitude:

$$P\left(+\frac{\hbar}{2}\right) = |\langle S_x; + | \hat{n}; + \rangle|^2,$$

Calculating the overlap:

$$\langle S_x; + | \hat{n}; + \rangle = \frac{1}{\sqrt{2}} [\langle + | + \rangle + \langle - | - \rangle] \left[\cos\left(\frac{\gamma}{2}\right) |+\rangle + \sin\left(\frac{\gamma}{2}\right) |-\rangle \right] = \frac{1}{\sqrt{2}} \left[\cos\left(\frac{\gamma}{2}\right) + \sin\left(\frac{\gamma}{2}\right) \right],$$

Squaring the amplitude:

$$P = \frac{1}{2} \left[\cos^2\left(\frac{\gamma}{2}\right) + \sin^2\left(\frac{\gamma}{2}\right) + 2 \cos\left(\frac{\gamma}{2}\right) \sin\left(\frac{\gamma}{2}\right) \right] = \frac{1}{2} [1 + \sin(\gamma)],$$

Expectation Value Using the operator $S_x = \frac{\hbar}{2} [|+\rangle\langle -| + |-\rangle\langle +|]$:

$$\langle S_x \rangle = \frac{\hbar}{2} \langle \hat{n}; + | [|+\rangle\langle -| + |-\rangle\langle +|] | \hat{n}; + \rangle,$$

$$\langle S_x \rangle = \frac{\hbar}{2} \left(2 \cos\left(\frac{\gamma}{2}\right) \sin\left(\frac{\gamma}{2}\right) \right) = \frac{\hbar}{2} \sin(\gamma),$$

For any spin-1/2 system, the square of a spin component operator is proportional to the identity: $S_i^2 = \frac{\hbar^2}{4} \mathbb{I}$. Thus:



$$\langle S_x^2 \rangle = \frac{\hbar^2}{4} \langle \hat{n}; + | \mathbb{I} | \hat{n}; + \rangle = \frac{\hbar^2}{4},$$

Dispersion Calculation:

$$\therefore \langle (S_x - \langle S_x \rangle)^2 \rangle = \frac{\hbar^2}{4} - \left(\frac{\hbar}{2} \sin(\gamma) \right)^2 = \frac{\hbar^2}{4} \cos^2(\gamma).$$

Summary of Special Cases:

$\gamma = 0$ (State is $|+\rangle$): $P = \frac{1}{2}$, $\langle (S_x - \langle S_x \rangle)^2 \rangle = \frac{\hbar^2}{4}$. (Maximal uncertainty, as S_z and S_x are incompatible).

$\gamma = \pi/2$ (State is $|S_x; +\rangle$): $P = 1$, $\langle (S_x - \langle S_x \rangle)^2 \rangle = 0$. (Zero uncertainty, as the state is an eigenstate of the measured observable).

$\gamma = \pi$ (State is $|-\rangle$): $P = \frac{1}{2}$, $\langle (S_x - \langle S_x \rangle)^2 \rangle = \frac{\hbar^2}{4}$.

Geometric Projection: The probability $P = \frac{1}{2} [1 + \sin(\gamma)]$ shows that as the state $|\hat{n}; +\rangle$ rotates from the z -axis ($\gamma = 0$) toward the x -axis ($\gamma = \pi/2$), the overlap with the S_x eigenstate increases continuously. This highlights how the Hilbert space geometry mirrors the physical orientation of the spin.

Quantum Fluctuation: The dispersion $\langle (S_x - \langle S_x \rangle)^2 \rangle = \frac{\hbar^2}{4} \cos^2(\gamma)$ quantifies the "quantum noise" inherent in measuring an observable that does not commute with the state's preparation.

At $\gamma = \pi/2$, the state is an eigenstate of S_x , and the dispersion vanishes, representing a state of definite polarization.

At $\gamma = 0$, the state is an eigenstate of S_z , which is maximally incompatible with S_x . Here, the dispersion reaches its maximum value $\left(\frac{\hbar^2}{4}\right)$, representing total uncertainty in the x -component.

This confirms that the variance of a spin component is strictly determined by its angular deviation from the axis of quantization.



1.15 A beam of spin $\frac{1}{2}$ atoms goes through a series of Stern–Gerlach type measurements as follows.

- The first measurement accepts $S_z = \hbar/2$ atoms and rejects $S_z = -\hbar/2$ atoms.
- The second measurement accepts $S_n = \hbar/2$ atoms and rejects $S_n = -\hbar/2$ atoms, where s_n is the eigenvalue of the operator $\vec{S} \cdot \hat{n}$, with $\hat{n}$ making an angle β in the xz -plane with respect to the z -axis.
- The third measurement accepts $S_z = -\hbar/2$ atoms and rejects $S_z = \hbar/2$ atoms.

What is the intensity of the final $S_z = -\hbar/2$ beam when the $S_z = \hbar/2$ beam surviving the first measurement is normalized to unity? How must we orient the second measuring apparatus if we are to maximize the intensity of the final $S_z = -\hbar/2$ beam?

Solution:

We analyze the beam intensity by tracking the state vector through each selective filter. Let the initial intensity after the first measurement be normalized to $I_0 = 1$.

First Measurement (S_z filter)

The apparatus accepts only $S_z = \hbar/2$. The state of the surviving atoms is:

$$|\Psi_1\rangle = |+\rangle,$$

Second Measurement (S_n filter)

The apparatus measures spin along $\hat{n}$, which lies in the xz -plane ($\alpha = 0$) at an angle β with respect to the z -axis. Following the general result derived in Problem 1.11, the eigenstate $|\hat{n}; +\rangle$ is:

$$|\hat{n}; +\rangle = \cos\left(\frac{\beta}{2}\right) |+\rangle + \sin\left(\frac{\beta}{2}\right) |-\rangle,$$

The probability of an atom passing through this second filter is:

$$P_2 = |\langle \hat{n}; + | \Psi_1 \rangle|^2 = \left| \langle + | \cos\left(\frac{\beta}{2}\right) | + \rangle \right|^2 = \cos^2\left(\frac{\beta}{2}\right),$$

The state of the atoms emerging from this stage is now $|\Psi_2\rangle = |\hat{n}; +\rangle$.

Step 3: Third Measurement (Selective $S_z = -\hbar/2$)

The final apparatus accepts only $S_z = -\hbar/2$, which corresponds to the state $|-\rangle$. The probability of passing this final stage, given it passed the second, is:

$$P_3 = |\langle - | \Psi_2 \rangle|^2 = \left| \langle - | \sin\left(\frac{\beta}{2}\right) | - \rangle \right|^2 = \sin^2\left(\frac{\beta}{2}\right),$$



Final Intensity Calculation

The total intensity I_{final} is the product of the successive probabilities:

$$I_{final} = I_0 \cdot P_2 \cdot P_3 = \cos^2\left(\frac{\beta}{2}\right) \sin^2\left(\frac{\beta}{2}\right),$$

Using the double-angle identity $\sin(\beta) = 2 \sin\left(\frac{\beta}{2}\right) \cos\left(\frac{\beta}{2}\right)$, we can simplify this to:

$$I_{final} = \frac{1}{4} \sin^2(\beta),$$

Optimization: Maximizing the Final Beam

To find the orientation β that maximizes I_{final} , we set the derivative with respect to β to zero:

$$\frac{d}{d\beta} \left[\frac{1}{4} \sin^2(\beta) \right] = \frac{1}{2} \sin(\beta) \cos(\beta) = \frac{1}{4} \sin(2\beta) = 0,$$

The maximum occurs when $\sin(2\beta) = 1$, (or by inspection of $\sin^2(\beta)$):

$$2\beta = \frac{\pi}{2} \Rightarrow \beta = \frac{\pi}{4} = 45^\circ,$$

Conclusion:

This result demonstrates the Quantum Eraser effect in its simplest form.

If $\beta = 0$ (second filter aligned with z), the intensity is zero because the state $|+\rangle$ has no $|-\rangle$ component.

By tilting the second apparatus to $\beta = \frac{\pi}{4} = 45^\circ$, we create a maximal "superposition" that re-introduces the $|-\rangle$ component with the highest efficiency possible for a three-filter setup, yielding a final intensity of 25% ($1/4$).



1.16 A certain observable in quantum mechanics has a 3×3 matrix representation as follows:

$$M = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

- Find the normalized eigenvectors of this observable and the corresponding eigenvalues. Is there any degeneracy?
- Give a physical example where all this is relevant.

Solution :

The given observable is:

$$M = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

Characteristic Equation:

To find the eigenvalues λ , we solve the secular equation $\det(M - \lambda \mathbb{I}) = 0$. Let $\lambda = \frac{\mu}{\sqrt{2}}$ to simplify the determinant:

$$\det \left(\frac{1}{\sqrt{2}} \begin{pmatrix} -\mu & 1 & 0 \\ 1 & -\mu & 1 \\ 0 & 1 & -\mu \end{pmatrix} \right) = 0,$$

Expanding along the first row:

$$-\mu(\mu^2 - 1) - 1(-\mu) = -\mu(\mu^2 - 2) = 0,$$

The roots for μ are 0 and $\pm\sqrt{2}$. Substituting back $\lambda = \frac{\mu}{\sqrt{2}}$, we obtain the eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = -1,$$

Degeneracy: Since all three eigenvalues are distinct, there is no degeneracy. The spectrum is simple.

Normalized Eigenvectors:

We solve the system $(M - \lambda \mathbb{I})|v_i\rangle = 0$ for each λ_i .

For $\lambda_1 = 1$ ($\mu = \sqrt{2}$):

The system yields $b = a\sqrt{2}$ and $c = a$. Normalizing $\langle v_1 | v_1 \rangle = a^2(1 + 2 + 1) = 4a^2 = 1 \Rightarrow a = 1/2$.

$$|1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix},$$



For $\lambda_2 = 0$ ($\mu = 0$):

The system yields $b = 0$ and $c = -a$. Normalizing $\langle v_2|v_2 \rangle = a^2(1 + 0 + 1) = 2a^2 = 1 \Rightarrow a = 1/\sqrt{2}$.

$$|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$$

For $\lambda_3 = -1$ ($\mu = -\sqrt{2}$):

The system yields $b = -a\sqrt{2}$ and $c = a$. Normalizing $\langle v_3|v_3 \rangle = a^2(1 + 2 + 1) = 4a^2 = 1 \Rightarrow a = 1/2$.

$$|-1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix},$$

b. Physical Example

This matrix represents the S_x operator for a spin-1 particle (in units of $\hbar$), expressed in the basis where S_z is diagonal (the standard $|1, m\rangle$ basis).

- Vector Mesons/Photons: Relevant in the study of particles with integer spin.
- Quadrupole Interactions: In nuclear physics, systems with $I = 1$ (like Deuterium) subjected to electric field gradients often utilize this matrix structure to describe transitions between magnetic sub-levels.
- Quantum Information: This defines a qutrit, the three-level generalization of a qubit, where the observable M would represent a measurement along a specific spatial axis.

Note that since M is a real symmetric matrix (and thus Hermitian), the eigenvectors are guaranteed to be orthonormal. You can verify that $\langle 1|0\rangle = 0$ and $\langle 1|-1\rangle = 0$ by direct inspection, confirming the consistency of the Hilbert space structure.

1.17 Let A and B be observables. Suppose the simultaneous eigenkets of A and B $\{|a', b'\rangle\}$ form a complete orthonormal set of base kets. Can we always conclude that :

$$[A, B] = 0?$$

If your answer is yes, prove the assertion. If your answer is no, give a counterexample.

Solution :

To prove that $[A, B] = 0$, we must show that the commutator acting on an arbitrary state $|\Psi\rangle$ results in the null vector, or equivalently, that the operator $AB - BA$ vanishes identically.



Action on a Basis Ket: Consider the action of the product AB on a specific simultaneous eigenket $|a', b'\rangle$:

$$AB|a', b'\rangle = A(B|a', b'\rangle) = A(b'|a', b'\rangle) = b'A|a', b'\rangle = b'a'|a', b'\rangle,$$

Since the eigenvalues a' and b' are real scalars (as A and B are observables), they commute:

$$AB|a', b'\rangle = a'b'|a', b'\rangle,$$

Symmetry of the Operation: Now, consider the action of the product BA on the same ket:

$$BA|a', b'\rangle = B(A|a', b'\rangle) = B(a'|a', b'\rangle) = a'B|a', b'\rangle = a'b'|a', b'\rangle,$$

The Commutator on the Basis: Subtracting the two results, we find:

$$[A, B]|a', b'\rangle = (AB - BA)|a', b'\rangle = (a'b' - a'b')|a', b'\rangle = 0,$$

Generalization via Completeness: Since the set $\{|a', b'\rangle\}$ is given to be complete, any arbitrary state $|\Psi\rangle$ in the Hilbert space can be expanded as:

$$|\Psi\rangle = \sum_{a', b'} c_{a', b'} [A, B]|a', b'\rangle = 0,$$

Since $[A, B]|\Psi\rangle = 0$ for all $|\Psi\rangle$, we conclude that $[A, B] = 0$.

It is important to distinguish this from the converse statement. While sharing a complete set of eigenkets implies $[A, B] = 0$, the existence of a commutator $[A, B] = 0$ only guarantees that there exists a common basis. If there is degeneracy, not every eigenket of A is necessarily an eigenket of B , but a common basis can always be constructed (e.g., via Gram-Schmidt within the degenerate subspaces). Here, the problem starts with the existence of the common basis as a given, making the proof direct.

1.18 Two Hermitian operators anticommute:

$$\{A, B\} = AB + BA = 0.$$

Is it possible to have a simultaneous (that is, common) eigenket of A and B ? Prove or illustrate your assertion.

Solution :

Suppose there exists a simultaneous eigenket $|\Psi\rangle = |a', b'\rangle$ such that:

$$A|\Psi\rangle = a'|\Psi\rangle,$$

$$B|\Psi\rangle = b'|\Psi\rangle,$$

We apply the anticommutator $\{A, B\} = AB + BA = 0$ to this state:

$$(AB + BA)|\Psi\rangle = 0,$$



Using the eigenvalue equations and the linearity of the operators:

$$A(B|\Psi\rangle) + B(A|\Psi\rangle) = 0,$$

$$A(b'|\Psi\rangle) + B(a'|\Psi\rangle) = 0,$$

$$b'A|\Psi\rangle + a'B|\Psi\rangle = 0,$$

Substituting the eigenvalues again:

$$b'a'|\Psi\rangle + a'b'|\Psi\rangle = 0,$$

$$2a'b'|\Psi\rangle = 0,$$

Since $|\Psi\rangle$ is an eigenket, it is by definition a non-null vector ($|\Psi\rangle \neq 0$). Therefore, the scalar coefficient must vanish:

$$a'b' = 0,$$

Illustration: The Spin- $\frac{1}{2}$ Case

Consider the spin operators S_x and S_y . We know they anticommute: $\{S_x, S_y\} = 0$. The eigenvalues for both are $\pm\hbar/2$. Since neither eigenvalue can be zero, it is impossible to find a common eigenket for S_x and S_y in a Spin- $\frac{1}{2}$ system.

However, consider a Spin-1 system where the eigenvalues are $\{\hbar, 0, -\hbar\}$. Here, a common eigenket could exist if its eigenvalue for at least one of the operators is 0.

In many physical systems, anticommuting operators are related to symmetries like Parity (P) and Charge Conjugation (C). If an operator O anticommutes with Parity (P), then O must map an even-parity state to an odd-parity state. The only way a state can be an eigenstate of both is if it sits in the "kernel" of the interaction, effectively having a null projection on the observable.



1.19 Two observables A_1 and A_2 , which do not involve time explicitly, are known not to commute,

$$[A_1, A_2] \neq 0,$$

yet we also know that A_1 and A_2 both commute with the Hamiltonian:

$$[A_1, H] = 0, \quad [A_2, H] = 0.$$

Prove that the energy eigenstates are, in general, degenerate. Are there exceptions? As an example, you may think of the central-force problem $H = \vec{p}^2/2m + V(r)$, with $A_1 \rightarrow L_z$, $A_2 \rightarrow L_x$.

Solution:

Assume, for the sake of contradiction, that an energy eigenvalue E is non-degenerate. This implies that the corresponding energy eigenspace is one-dimensional, spanned by a single normalized ket $|n\rangle$ such that $H|n\rangle = E|n\rangle$.

Constants of Motion: Since $[A_1, H] = 0$, the operator A_1 preserves the energy eigenspace.

Acting with H on the state $A_1|n\rangle$:

$$H(A_1|n\rangle) = A_1H|n\rangle = A_1(E|n\rangle) = E(A_1|n\rangle),$$

Thus, $A_1|n\rangle$ is also an energy eigenket with the same eigenvalue E .

Implication of Non-Degeneracy: In a non-degenerate subspace, any vector with energy E must be proportional to $|n\rangle$. Therefore, $|n\rangle$ must be an eigenket of A_1 :

$$A_1|n\rangle = \lambda_1|n\rangle,$$

Simultaneous Eigenkets: Following the same logic for A_2 (since $[A_2, H] = 0$):

$$A_2|n\rangle = \lambda_2|n\rangle,$$

The Contradiction: If the energy levels were non-degenerate, the energy eigenkets would form a basis of simultaneous eigenkets for A_1 and A_2 . According to the theorem proven in Problem 1.17, the existence of such a basis implies that the operators must commute ($[A_1, A_2] = 0$).

However, we are given that $[A_1, A_2] \neq 0$. Therefore, our initial assumption must be false: the energy eigenstates must, in general, be degenerate.

Are there exceptions? Yes. An exception occurs if a specific state $|n\rangle$ is an eigenstate of both A_1 and A_2 despite them not commuting globally.



This usually happens when the state is a singlet of the symmetry group (i.e., it is invariant under the transformations generated by A_1 and A_2). Mathematically, this occurs if:

$$A_1|n\rangle = c_1|n\rangle, \quad \text{and} \quad A_2|n\rangle = c_2|n\rangle,$$

Typically, these eigenvalues are zero. If a subspace is one-dimensional and satisfies this, it can remain non-degenerate.

Example: Central-Force Problem

In the Hamiltonian $H = \vec{p}^2/2m + V(r)$:

Symmetries: The system is spherically symmetric, so $[L_z, H] = 0$ and $[L_x, H] = 0$.

Non-Commutativity: $[L_z, L_x] = i\hbar L_y \neq 0$.

General Case (Degenerate): For any orbital angular momentum $l > 0$, the energy levels are $(2l + 1)$ -degenerate. L_z and L_x act within this subspace, mapping states into one another without changing the energy.

Exception (Non-Degenerate): The ground state ($l = 0$). For the s-wave state $|0,0\rangle$ we have:

$$L_z|0,0\rangle = 0, \quad \text{and} \quad L_x|0,0\rangle = 0,$$

Since it is annihilated by both, it is a simultaneous eigenstate with eigenvalue zero, allowing it to be non-degenerate.

From the perspective of Group Theory, A_1 and A_2 are generators of a non-Abelian Lie Algebra. The Hilbert space decomposes into irreducible representations (irreps). If the algebra is non-Abelian, it must contain irreps with dimension $d > 1$. Each such irrep corresponds to a degenerate energy level.

1.20 a. The simplest way to derive the Schwarz inequality goes as follows. First, observe:

$$(\langle\alpha| + \lambda^*\langle\beta|) \cdot (|\alpha\rangle + \lambda|\beta\rangle) > 0,$$

for any complex number λ ; then choose λ in such a way that the preceding inequality reduces to the Schwarz inequality.

b. Show that the equality sign in the generalized uncertainty relation holds if the state in question satisfies:

$$\Delta A|\alpha\rangle = \lambda\Delta B|\alpha\rangle,$$

With λ purely imaginary.



c. Explicit calculations using the usual rules of wave mechanics show that the wave function for a Gaussian wave packet given by:

$$\langle x'|\alpha\rangle = (2\pi d^2)^{-\frac{1}{4}} \exp\left[\frac{i\langle p\rangle x'}{\hbar} - \frac{(x' - \langle x\rangle)^2}{4d^2}\right],$$

satisfies the minimum uncertainty relation:

$$\sqrt{\langle(\Delta x)^2\rangle}\sqrt{\langle(\Delta p)^2\rangle} = \frac{\hbar}{2},$$

Prove that the requirement:

$$\langle x'|\Delta x|\alpha\rangle = (\text{imaginary number})\langle x'|\Delta p|\alpha\rangle,$$

is indeed satisfied for such a Gaussian wave packet, in agreement with (b).

Solution:

a. Derivation of the Schwarz Inequality

We start with the fundamental axiom that the norm of any state vector in Hilbert space is non-negative. Define a ket $|\Psi\rangle = |\alpha\rangle + \lambda|\beta\rangle$, where λ is a complex scalar.

$$\langle\Psi|\Psi\rangle = (\langle\alpha| + \lambda^*\langle\beta|)(|\alpha\rangle + \lambda|\beta\rangle) \geq 0,$$

Expanding the terms:

$$\langle\alpha|\alpha\rangle + \lambda^*\langle\beta|\alpha\rangle + \lambda\langle\alpha|\beta\rangle + |\lambda|^2\langle\beta|\beta\rangle \geq 0,$$

To find the most restrictive condition (the inequality's bound), we choose λ such that it "minimizes" the expression. Setting $\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}$, $\lambda^* = -\frac{\langle\alpha|\beta\rangle}{\langle\beta|\beta\rangle}$:

$$\langle\alpha|\alpha\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\beta|\beta\rangle}\langle\beta|\alpha\rangle - \frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}\langle\alpha|\beta\rangle + \frac{|\langle\beta|\alpha\rangle|^2}{\langle\beta|\beta\rangle^2}\langle\beta|\beta\rangle \geq 0,$$

Using $\langle\alpha|\beta\rangle\langle\beta|\alpha\rangle = |\langle\beta|\alpha\rangle|^2$:

$$\langle\alpha|\alpha\rangle - \frac{|\langle\beta|\alpha\rangle|^2}{\langle\beta|\beta\rangle} \geq 0 \Rightarrow \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle \geq |\langle\beta|\alpha\rangle|^2,$$

b. Equality in the Generalized Uncertainty Relation

The generalized uncertainty relation is derived using the Schwarz inequality for the kets $|f\rangle = \Delta A|\alpha\rangle$ and $|g\rangle = \Delta B|\alpha\rangle$. Equality holds if and only if two conditions are met:

$$\langle(\Delta A)^2\rangle\langle(\Delta B)^2\rangle \geq |\langle\Delta A\Delta B\rangle|^2,$$

Linear dependence (Schwarz equality): $|f\rangle = \lambda|g\rangle$. This implies:

$$\Delta A|\alpha\rangle = \lambda\Delta B|\alpha\rangle,$$



Vanishing of the Anticommutator: The uncertainty relation actually states $\langle(\Delta A)^2\rangle\langle(\Delta B)^2\rangle \geq \frac{1}{4}|\langle\{A, B\}\rangle + \langle[A, B]\rangle|^2$. For the lower bound to be exactly the commutator term, we require $\{\Delta A, \Delta B\} = 0$.

If $\Delta A|\alpha\rangle = \lambda\Delta B|\alpha\rangle$, then:

$$\langle\{\Delta A, \Delta B\}\rangle = \langle\alpha|\Delta A\Delta B + \Delta B\Delta A|\alpha\rangle = (\lambda + \lambda^*)\langle\alpha|\Delta B^2|\alpha\rangle,$$

This vanishes if $\lambda + \lambda^* = 0$, meaning λ must be purely imaginary.

c. Verification for the Gaussian Wave Packet

We evaluate the action of $\Delta x = x - \langle x \rangle$ and $\Delta p = p - \langle p \rangle$ on the given wave function $\Psi(x') = \langle x'|\alpha\rangle$.

Left-Hand Side (LHS):

$$\langle x'|\Delta x|\alpha\rangle = (x' - \langle x \rangle)\Psi(x'),$$

Right-Hand Side (RHS):

Recall that $p = -i\hbar \frac{\partial}{\partial x'}$. The derivative of the Gaussian is:

$$\frac{\partial \Psi}{\partial x'} = \left[\frac{i\langle p \rangle}{\hbar} - \frac{2(x' - \langle x \rangle)}{4d^2} \right] \Psi(x'),$$

Thus, Δp acting on Ψ is:

$$\langle x'|\Delta p|\alpha\rangle = -i\hbar \frac{\partial \Psi}{\partial x'} - \langle p \rangle \Psi = \left[\langle p \rangle + \frac{i\hbar(x' - \langle x \rangle)}{2d^2} \right] \Psi - \langle p \rangle \Psi,$$

$$\langle x'|\Delta p|\alpha\rangle = \left[\frac{i\hbar(x' - \langle x \rangle)}{2d^2} \right] \Psi$$

Comparison:

$$\langle x'|\Delta x|\alpha\rangle = (x' - \langle x \rangle)\Psi(x') = \left(\frac{2d^2}{i\hbar} \right) \left[\frac{i\hbar(x' - \langle x \rangle)}{2d^2} \right] \Psi,$$

$$\langle x'|\Delta x|\alpha\rangle = \frac{-i2d^2}{\hbar} \langle x'|\Delta p|\alpha\rangle,$$

Since $\lambda = -i\frac{2d^2}{\hbar}$ is purely imaginary, the condition for minimum uncertainty is satisfied.

$$\langle(\Delta x)^2\rangle\langle(\Delta p)^2\rangle = \frac{1}{4}|\langle\{x, p\}\rangle + \langle[x, p]\rangle|^2,$$

$$\sqrt{\langle(\Delta x)^2\rangle}\sqrt{\langle(\Delta p)^2\rangle} = \frac{\hbar}{2},$$



This Gaussian state is the "most classical" of all quantum states. In the language of quantum optics, it is a coherent state (if $d^2 = \hbar/2m\omega$). The fact that λ is constant across all x' ensures that the "shape" of the uncertainty is perfectly balanced between position and momentum.

1.21 a. Compute

$$\langle (\Delta S_x)^2 \rangle \equiv \langle S_x^2 \rangle - \langle S_x \rangle^2,$$

where the expectation value is taken for the $S_z +$ state. Using your result, check the generalized uncertainty relation:

$$\langle \Delta A \rangle^2 \langle \Delta B \rangle^2 \geq \frac{1}{4} |\langle [A, B] \rangle|^2,$$

With $A \rightarrow S_x, B \rightarrow S_y$.

b. Check the uncertainty relation with $A \rightarrow S_x, B \rightarrow S_y$ for the $S_x +$ state.

Solution:

a. Uncertainty in the $S_z +$ state ($|+\rangle$):

Computing $\langle (\Delta S_x)^2 \rangle$:

We use the expectation values for the state $|\Psi\rangle = |+\rangle$:

$$\text{First Moment: } \langle S_x \rangle = \left\langle + \left| \frac{\hbar}{2} (|+\rangle\langle -| + |-\rangle\langle +|) \right| + \right\rangle = 0,$$

Second Moment: Since $S_x^2 = \frac{\hbar^2}{4} \mathbb{I}$ for spin-1/2:

$$\langle S_x^2 \rangle = \frac{\hbar^2}{4} \langle + | \mathbb{I} | + \rangle = \frac{\hbar^2}{4},$$

$$\text{Variance: } \langle (\Delta S_x)^2 \rangle = \frac{\hbar^2}{4} + 0 = \frac{\hbar^2}{4}.$$

Checking the Generalized Uncertainty Relation:

$$\langle \Delta S_x \rangle^2 \langle \Delta S_y \rangle^2 \geq \frac{1}{4} |\langle [S_x, S_y] \rangle|^2,$$

By symmetry, $\langle (\Delta S_x)^2 \rangle$ in the $|+\rangle$ state is also $\frac{\hbar^2}{4}$.

$$\langle \Delta S_x \rangle^2 \langle \Delta S_y \rangle^2 = \frac{\hbar^2}{4} \frac{\hbar^2}{4} = \frac{\hbar^4}{16},$$

Recall the fundamental commutator $[S_x, S_y] = i\hbar S_z$.

$$\langle [S_x, S_y] \rangle = i\hbar \langle + | S_z | + \rangle = i\hbar \frac{\hbar}{2} = \frac{i\hbar^2}{2},$$



$$\frac{1}{4} |\langle [S_x, S_y] \rangle|^2 = \frac{1}{4} \left| \frac{i\hbar^2}{2} \right|^2 = \frac{\hbar^4}{16},$$

The $|+\rangle$ state is an intelligent state (or minimum uncertainty state) for the S_x and S_y observables.

- b. Uncertainty in the $S_x +$ state ($|S_x; +\rangle$):

For the state $|S_x; +\rangle$, the operator S_x has a definite value ($\hbar/2$). Thus, the dispersion $\langle (\Delta S_x)^2 \rangle = 0$.

The dispersion $\langle (\Delta S_y)^2 \rangle$ will be non-zero (specifically $\hbar^2/4$), but the product is:

$$\langle \Delta S_x \rangle^2 \langle \Delta S_y \rangle^2 = (0) \frac{\hbar^2}{4} = 0,$$

We evaluate the expectation value of the commutator in the $|S_x; +\rangle$ state:

$$\langle [S_x, S_y] \rangle = i\hbar \langle S_x; + | S_z | S_x; + \rangle,$$

Using $|S_x; +\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$:

$$\begin{aligned} \langle [S_x, S_y] \rangle &= \frac{i\hbar}{2} (\langle + | + \langle - |) S_z (|+\rangle + |-\rangle) = \frac{i\hbar}{2} \left[\frac{\hbar}{2} - \frac{\hbar}{2} \right] = 0, \\ \frac{1}{4} |\langle [S_x, S_y] \rangle|^2 &= \frac{1}{4} |0|^2 = 0. \end{aligned}$$

In part (b), we find $0 \geq 0$. This is a "trivial" satisfaction of the uncertainty principle. It highlights a critical conceptual point: the uncertainty relation does not force uncertainty if the state is an eigenstate of the commutator's resulting operator. Since $|S_x; +\rangle$ is "unpolarized" in the z -direction ($\langle S_z \rangle = 0$), the geometric requirement for non-zero product uncertainty disappears.

1.22 Find the linear combination of $|+\rangle$ and $|-\rangle$ kets that maximizes the uncertainty product:

$$\langle (\Delta S_x)^2 \rangle \langle (\Delta S_y)^2 \rangle.$$

Verify explicitly that for the linear combination you found, the uncertainty relation for S_x and S_y is not violated.

Solution:

We seek to find a general state $|\Psi\rangle$ that maximizes the product of variances $P = \langle (\Delta S_x)^2 \rangle \langle (\Delta S_y)^2 \rangle$.



Any state in the spin-1/2 Hilbert space can be represented as a point on the Bloch sphere with unit vector $\hat{n} = (n_x, n_y, n_z)$:

$$|\Psi\rangle = \cos\left(\frac{\beta}{2}\right)|+\rangle + e^{i\alpha} \sin\left(\frac{\beta}{2}\right)|-\rangle,$$

The expectation values are $\langle S_i \rangle = \frac{\hbar}{2} n_i$. Specifically:

$$\langle S_x \rangle = \frac{\hbar}{2} \sin(\beta) \cos(\alpha), \quad \langle S_y \rangle = \frac{\hbar}{2} \sin(\beta) \sin(\alpha), \quad \langle S_z \rangle = \frac{\hbar}{2} \cos(\beta),$$

Calculation of the Uncertainty Product

Recall that for spin-1/2, $S_i^2 = \frac{\hbar^2}{4} \mathbb{I}$, so the variance is $\langle (\Delta S_i)^2 \rangle = \frac{\hbar^2}{4} - \langle S_i \rangle^2 = \frac{\hbar^2}{4} (1 - n_i^2)$.

The product is:

$$P = \left(\frac{\hbar^2}{4}\right)^2 (1 - n_x^2)(1 - n_y^2) = \frac{\hbar^4}{16} (1 - (n_x^2 + n_y^2) + n_z^2 n_y^2),$$

Using the constraint for a unit vector $n_x^2 + n_y^2 = 1 - n_z^2$:

$$P = \frac{\hbar^4}{16} (1 - (1 - n_z^2) + n_z^2 n_y^2) = \frac{\hbar^4}{16} (n_z^2 + n_z^2 n_y^2),$$

To maximize $n_z^2 + n_z^2 n_y^2$, we consider two cases:

1. At the equator ($n_z = 0$): $P = \frac{\hbar^4}{16} (n_z^2 n_y^2)$. The maximum value of $x^2 y^2$ subject to $x^2 + y^2 = 1$ is $1/4$ (at $x^2 = y^2 = 1/2$).
2. At the poles ($n_z^2 = 1$): Here $n_x = n_y = 0$, so $P = \frac{\hbar^4}{16} (1 + 0) = \frac{\hbar^4}{16}$.

Comparing the values, the maximum occurs when $n_z = \pm 1$. Therefore, the linear combinations that maximize the product are simply the eigenstates of S_z :

$$|\Psi\rangle = |+\rangle, \quad |\Psi\rangle = |-\rangle,$$

The generalized uncertainty relation for S_x and S_y is:

$$\langle (\Delta S_x)^2 \rangle \langle (\Delta S_y)^2 \rangle \geq \frac{1}{4} |\langle [S_x, S_y] \rangle|^2,$$

For the state $|+\rangle$:

$$\langle (\Delta S_x)^2 \rangle \langle (\Delta S_y)^2 \rangle = \frac{\hbar^4}{16},$$

Using $[S_x, S_y] = i\hbar S_z$:

$$\frac{1}{4} |\langle [S_x, S_y] \rangle|^2 = \frac{1}{4} |i\hbar \langle +|S_z|+\rangle|^2 = \frac{\hbar^2}{4} \left|\frac{\hbar}{2}\right|^2 = \frac{\hbar^4}{16},$$



The uncertainty relation is satisfied as an equality. This proves that $|+\rangle$ is a minimum uncertainty state (or intelligent state) for S_x and S_y , even though it maximizes the value of the product compared to other states like $|S_x; +\rangle$ where the product would be zero.

1.23 Evaluate the x - p uncertainty product $\langle(\Delta x)^2\rangle\langle(\Delta p)^2\rangle$, for a one-dimensional particle confined between two rigid walls:

$$V = \begin{cases} 0 & \text{for } 0 < x < a, \\ \infty & \text{otherwise.} \end{cases}$$

Do this for both the ground and excited states.

Solution:

The particle is confined in a potential $V(x) = 0$ for $0 < x < a$, and $V(x) = \infty$ elsewhere.

Inside the well, the time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m} \frac{d^2\Psi}{dx^2} = E\Psi \Rightarrow \frac{d^2\Psi}{dx^2} + k^2\Psi = 0, \quad k = \frac{\sqrt{2mE}}{\hbar},$$

The general solution for the harmonic oscillator equation is:

$$\Psi(x) = A \sin(kx) + B \cos(kx),$$

The wave function must vanish at the rigid walls due to the infinite potential:

Condition at $x = 0$: $\Psi(0) = B \cos(0) = 0 \Rightarrow B = 0$.

Condition at $x = a$: $\Psi(a) = A \sin(ka) = 0$ For no trivial solution ($A \neq 0$), we require $ka = n\pi$, where $n = 1, 2, 3, \dots$

Thus, the allowed wavenumbers are $k_n = \frac{n\pi}{a}$.

The probability of finding the particle within the box must be unity:

$$\int_0^a |\Psi_n(x)|^2 dx = |A|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{|A|^2}{2} \int_0^a \left(1 - \cos\left(2\frac{n\pi x}{a}\right)\right) dx = |A|^2 \left(\frac{a}{2}\right) = 1,$$

Solving for A (choosing the phase to be real): $A = \sqrt{\frac{2}{a}}$.

Final Eigenfunctions:

$$\Psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right),$$

Due to the symmetry of the probability density $\Psi_n(x)$ about the center of the well, the expectation value of the position is:



$$\langle x \rangle = \frac{a}{2},$$

The second moment $\langle x^2 \rangle$ is calculated as follows:

$$\langle x^2 \rangle = \int_0^a x^2 |\Psi_n(x)|^2 dx = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx,$$

Using the identity $\sin^2(\theta) = \frac{1}{2}[1 - \cos(2\theta)]$ and performing integration by parts:

$$\begin{aligned} \langle x^2 \rangle &= \frac{1}{a} \left[\int_0^a x^2 dx + \int_0^a x^2 \cos\left(\frac{2n\pi x}{a}\right) dx \right], \\ \langle x^2 \rangle &= \frac{1}{a} \left[\frac{a^3}{3} + a^2(0) - \frac{a}{n\pi} \int_0^a x \sin\left(\frac{2n\pi x}{a}\right) dx \right], \\ \langle x^2 \rangle &= \frac{1}{a} \left[\frac{a^3}{3} - \frac{a}{n\pi} \left[\frac{a^2}{2n\pi} \cos(2n\pi) - \int_0^a \cos\left(\frac{2n\pi x}{a}\right) dx \right] \right], \\ \langle x^2 \rangle &= \frac{1}{a} \left[\frac{a^3}{3} - \frac{a}{n\pi} \left[\frac{a^2}{2n\pi} \right] \right] = a^2 \left[\frac{1}{3} - \frac{1}{2n^2\pi^2} \right], \end{aligned}$$

Thus, the position variance is:

$$\langle (\Delta x)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2 = a^2 \left[\frac{1}{3} - \frac{1}{2n^2\pi^2} \right] - \frac{a^2}{4} = a^2 \left[\frac{1}{12} - \frac{1}{2n^2\pi^2} \right],$$

Since the wave function is stationary and real, there is no net flux, meaning:

$$\langle p \rangle = 0,$$

The second moment $\langle p^2 \rangle$ is directly related to the energy eigenvalues $E_n = \frac{p^2}{2m}$:

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} \Rightarrow \langle p^2 \rangle = 2mE_n = \frac{n^2\pi^2\hbar^2}{a^2},$$

Consequently, $\langle (\Delta p)^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}$.

Multiplying the two dispersions for the n -th state:

$$\begin{aligned} \langle (\Delta x)^2 \rangle \langle (\Delta p)^2 \rangle &= a^2 \left[\frac{1}{12} - \frac{1}{2n^2\pi^2} \right] \frac{n^2\pi^2\hbar^2}{a^2}, \\ \langle (\Delta x)^2 \rangle \langle (\Delta p)^2 \rangle &= \hbar^2 \left[\frac{n^2\pi^2}{12} - \frac{1}{2} \right], \end{aligned}$$



Evaluation for Ground and Excited States

STATE	n	$\langle(\Delta x)^2\rangle\langle(\Delta p)^2\rangle$	NUMERICAL VALUE
GROUND STATE	1	$\hbar^2 \left[\frac{\pi^2}{12} - \frac{1}{2} \right]$	$\approx 0.322\hbar^2$
1ST EXCITED	2	$\hbar^2 \left[\frac{\pi^2}{3} - \frac{1}{2} \right]$	$\approx 2.790\hbar^2$
2ND EXCITED	3	$\hbar^2 \left[\frac{3\pi^2}{4} - \frac{1}{2} \right]$	$\approx 6.902\hbar^2$

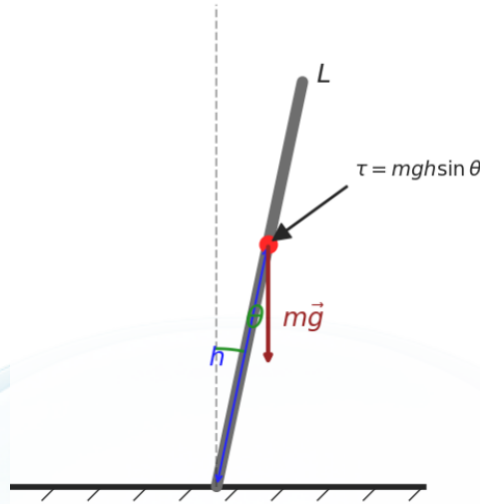
The ground state uncertainty product ($\approx 0.322\hbar^2$) is strictly greater than the Heisenberg lower bound of $0.25\hbar^2$. This is expected because the infinite well does not support the Gaussian wave packet required for minimum uncertainty.

As $n \rightarrow \infty$, the uncertainty product grows quadratically. Physically, this represents the transition to the classical limit: while the spatial distribution becomes uniform ($\langle(\Delta x)^2\rangle \rightarrow \frac{a^2}{12}$), the kinetic energy (and thus momentum dispersion) increases without bound to maintain the high-frequency oscillations of the wave function.

1.24 Estimate the rough order of magnitude of the length of time that an ice pick can be balanced on its point if the only limitation is that set by the Heisenberg uncertainty principle. Assume that the point is sharp and that the point and the surface on which it rests are hard. You may make approximations which do not alter the general order of magnitude of the result. Assume reasonable values for the dimensions and weight of the ice pick. Obtain an approximate numerical result and express it in seconds.



Solution :



We model the ice pick as a rigid rod of length L and mass m , balanced on a sharp point. Let θ be the angle from the vertical. For small angles, the equation of motion is:

$$I\ddot{\theta} = \tau = mgh \sin(\theta) \approx mgh\theta,$$

Where I is the moment of inertia about the tip and h is the distance from the tip to the center of mass. Defining $\omega^2 = \frac{mgh}{I}$, the general solution is:

$$\theta(t) = \theta_0 \cosh(\omega t) + \frac{\dot{\theta}_0}{\omega} \sinh(\omega t) \approx \frac{1}{2} \left(\theta_0 + \frac{\dot{\theta}_0}{\omega} \right) e^{\omega t},$$

where we consider only the growing exponential term for large t .

According to the Heisenberg Uncertainty Principle, the initial position and momentum cannot both be zero. In angular variables:

$$\Delta\theta\Delta L_\theta \geq \frac{\hbar}{2} \Rightarrow \theta_0(I\dot{\theta}_0) \sim \frac{\hbar}{2},$$

To maximize the time t , we must minimize the initial "kick" coefficient $A = \theta_0 + \frac{\dot{\theta}_0}{\omega}$.

Substituting $\dot{\theta}_0 = \frac{\hbar}{2I\theta_0}$:

$$A(\theta_0) = \theta_0 + \frac{\hbar}{2I\omega\theta_0},$$

Minimizing A with respect to θ_0 ($\frac{dA}{d\theta_0} = 0$) gives :

$$1 - \frac{\hbar}{2I\omega\theta_0^2} = 0,$$



$$\theta_0 = \sqrt{\frac{\hbar}{2I\omega}} \Rightarrow A_{min} = \sqrt{\frac{2\hbar}{I\omega'}}$$

We assume reasonable values for an ice pick:

Length (L): $\approx 15\text{cm} = 0.15\text{m}$.

Mass(m): $\approx 100\text{g} = 0.1\text{Kg}$.

Geometry: Assuming a uniform rod, $h = \frac{L}{2} = 0.075\text{m}$ and $I = \frac{1}{3}mL^2 = 7.5 \times 10^{-4}\text{Kg} \cdot \text{m}^2$.

Characteristic frequency: $\omega = \sqrt{\frac{3g}{2L}} = \sqrt{\frac{3(9.8)}{0.3}}\text{s}^{-1} \approx 9.9\text{s}^{-1}$.

The ice pick "falls" when the angle reaches the order of $\theta(t) \sim 1$ radian:

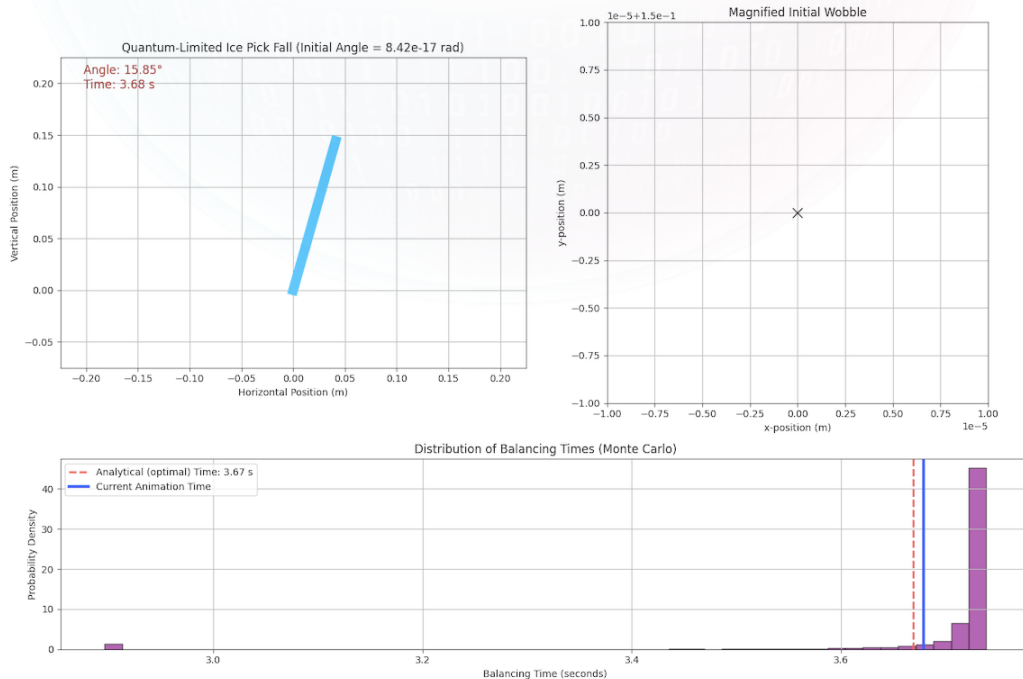
$$1 \approx \sqrt{\frac{2\hbar}{I\omega}} e^{\omega t} \Rightarrow t \approx \frac{1}{2\omega} \ln\left(\frac{I\omega}{2\hbar}\right),$$

Plugging in the constants ($\hbar \approx 1.054 \times 10^{-34} \text{J} \cdot \text{s}$):

$$t \approx \frac{1}{2(9.9\text{s}^{-1})} \ln\left(\frac{(7.5 \times 10^{-4}\text{Kg} \cdot \text{m}^2)(9.9\text{s}^{-1})}{2(1.054 \times 10^{-34} \text{J} \cdot \text{s})}\right),$$

$$t \approx \frac{1}{19.8\text{s}^{-1}} \ln(3.5 \times 10^{31}) \approx \frac{72.6}{19.8} \text{s} \approx 3.7\text{s}.$$

The rough order of magnitude for balancing an ice pick under purely quantum limitations is approximately 3 to 4 seconds. Despite the tiny scale of $\hbar$, the exponential growth of the instability ($e^{\omega t}$) amplifies quantum uncertainties to macroscopic scales very rapidly.





1.25 Consider a three-dimensional ket space. If a certain set of orthonormal kets, say, $|1\rangle$, $|2\rangle$, and $|3\rangle$, are used as the base kets, the operators A and B are represented by:

$$A \doteq \begin{pmatrix} a & 0 & 0 \\ 0 & -a & 0 \\ 0 & 0 & -a \end{pmatrix}, \quad B \doteq \begin{pmatrix} b & 0 & 0 \\ 0 & 0 & -ib \\ 0 & ib & 0 \end{pmatrix},$$

with a and b both real.

- Obviously A exhibits a degenerate spectrum. Does B also exhibit a degenerate spectrum?
- Show that A and B commute.
- Find a new set of orthonormal kets which are simultaneous eigenkets of both A and B . Specify the eigenvalues of A and B for each of the three eigenkets. Does your specification of eigenvalues completely characterize each eigenket?

Solution:

- The eigenvalues λ of the operator B are the roots of the characteristic equation $\det(B - \lambda I) = 0$.

The characteristic polynomial is:

$$\det \begin{pmatrix} b - \lambda & 0 & 0 \\ 0 & -\lambda & -ib \\ 0 & ib & -\lambda \end{pmatrix} = (b - \lambda) \det \begin{pmatrix} -\lambda & -ib \\ ib & -\lambda \end{pmatrix} = (b - \lambda)(\lambda^2 - b^2),$$

$$0 = -(\lambda - b)(\lambda - b)(\lambda + b),$$

The eigenvalues are $\lambda_1 = b$, $\lambda_2 = b$, and $\lambda_3 = -b$.

Since the eigenvalue $\lambda = b$ has an algebraic multiplicity of 2, the spectrum of B is degenerate.

- We compute the commutator directly via matrix multiplication:

$$AB = \begin{pmatrix} a & 0 & 0 \\ 0 & -a & 0 \\ 0 & 0 & -a \end{pmatrix} \begin{pmatrix} b & 0 & 0 \\ 0 & 0 & -ib \\ 0 & ib & 0 \end{pmatrix} = \begin{pmatrix} ab & 0 & 0 \\ 0 & 0 & iab \\ 0 & -iab & 0 \end{pmatrix},$$

$$BA = \begin{pmatrix} b & 0 & 0 \\ 0 & 0 & -ib \\ 0 & ib & 0 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & -a & 0 \\ 0 & 0 & -a \end{pmatrix} = \begin{pmatrix} ab & 0 & 0 \\ 0 & 0 & iab \\ 0 & -iab & 0 \end{pmatrix},$$

Observing that $AB = BA$, it follows that $[A, B] = 0$. This implies that A and B share a simultaneous eigenbasis.



- c. Since $|1\rangle$, is an eigenket of A with a non-degenerate eigenvalue a , it must be an eigenket of B .

$$|\Psi_1\rangle = |1\rangle \Rightarrow \{a, b\},$$

We seek linear combinations $|\pm\rangle = \alpha|2\rangle + \beta|3\rangle$ that diagonalize B :

$$B(\alpha|2\rangle + \beta|3\rangle) = \lambda(\alpha|2\rangle + \beta|3\rangle),$$

Substituting the operator definition: $-ib\alpha|2\rangle + ib\beta|3\rangle = \lambda\alpha|2\rangle + \lambda\beta|3\rangle$.

Equating coefficients: $ib\beta = \lambda\alpha$ and $-ib\alpha = \lambda\beta$.

For $\lambda = b$ and $\alpha = i\beta$. Normalizing, we get:

$$|\Psi_2\rangle = \frac{1}{\sqrt{2}}[i|2\rangle + |3\rangle] \Rightarrow \{-a, b\},$$

For $\lambda = -b$ and $\alpha = -i\beta$. Normalizing, we get:

$$|\Psi_3\rangle = \frac{1}{\sqrt{2}}[-i|2\rangle + |3\rangle] \Rightarrow \{-a, -b\},$$

Characterization: The state of the system is completely determined by the pair of eigenvalues (a_i, b_i) . Even though A and B are individually degenerate, the mapping $i \rightarrow (a_i, b_i)$ is injective:

$$\text{spec}(A, B) = \{(a, b), (-a, b), (-a, -b)\},$$

Since no two eigenkets share the same set of eigenvalues, $\{A, B\}$ constitutes a Complete Set of Commuting Observables (CSCO).



1.26 a. Prove that $(1/\sqrt{2})(1 + i\sigma_x)$, where the matrix σ_x is :

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

acting on a two-component spinor can be regarded as the matrix representation of the rotation operator about the x -axis by angle $-\pi/2$. (The minus sign signifies that the rotation is clockwise.)

b. Construct the matrix representation of S_z when the eigenkets of S_y are used as base vectors.

Solution:

a. Unitary Representation of Rotations

In the context of a spin-1/2 system, the rotation operator for a rotation by an angle ϕ about an axis defined by the unit vector $\hat{n}$ is given by the exponential map:

$$D(\hat{n}, \phi) = \exp\left(-i \frac{\vec{S} \cdot \hat{n}}{\hbar} \phi\right) = \exp\left(-i \frac{\vec{\sigma} \cdot \hat{n}}{2} \phi\right),$$

For a rotation about the x -axis ($\hat{n} = \hat{e}_x$), the operator simplifies to:

$$D_x(\phi) = \exp\left(-i \frac{\sigma_x}{2} \phi\right),$$

Using the fundamental property of Pauli matrices ($\sigma_k^2 = \mathbb{I}$), the Taylor expansion of the exponential yields the Euler-like identity:

$$\exp(i\sigma_k\theta) = \mathbb{I} \cos(\theta) + i\sigma_k \sin(\theta),$$

Setting the angle $\phi = -\frac{\pi}{2}$, we define the argument $\theta = -\frac{\phi}{2} = \frac{\pi}{4}$. Substituting this into the identity:

$$D_x(-\pi/2) = \mathbb{I} \cos\left(\frac{\pi}{4}\right) + i\sigma_x \sin\left(\frac{\pi}{4}\right),$$

Since $\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = 1/\sqrt{2}$, we obtain:

$$D_x(-\pi/2) = \frac{1}{\sqrt{2}}(\mathbb{I} + i\sigma_x),$$

The operator given is precisely the unitary matrix representing a clockwise rotation of $\pi/2$ radians in the spinor space.



b. Change of Basis for the S_z Operator

Let $\{|y, +\rangle, |y, -\rangle\}$ be the orthonormal basis of S_y eigenkets. In the standard S_z basis $\{|+\rangle, |-\rangle\}$, these are defined as:

$$|y, \pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm i|-\rangle),$$

The matrix elements of S_z in the S_y basis are given by $M_{ij} = \langle y, i | S_z | y, j \rangle$. Recalling that $S_z = \frac{\hbar}{2}(|+\rangle\langle+| - |-\rangle\langle-|)$, Let's use the transformation matrix U where $|y, \pm\rangle$ are columns:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \quad U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix},$$

The representation is $S_z^{(y)} = U^\dagger S_z^{(z)} U$:

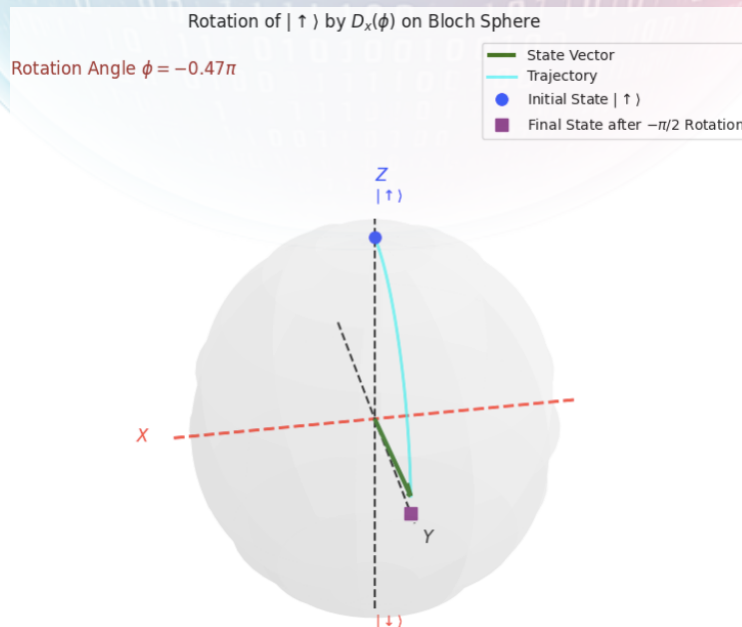
$$S_z^{(y)} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix},$$

$$S_z^{(y)} = \frac{\hbar}{4} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix},$$

In the S_y basis, the operator S_z is represented by the matrix:

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2} \sigma_x,$$

This result is consistent with the cyclic permutation of indices in the $SO(3)$ symmetry of spin space.





1.27 Some authors define an operator to be real when every member of its matrix elements $\langle b'|A|b'' \rangle$ is real in some representation ($\{|b'\rangle\}$ basis in this case). Is this concept representation independent, that is, do the matrix elements remain real even if some basis other than $\{|b'\rangle\}$ is used? Check your assertion using familiar operators such as S_y and S_z (see Problem 1.26) or x and p_x .

Solution:

Given the definition that an operator A is "real" if its matrix elements $\langle b'|A|b'' \rangle$ are real in a specific basis $\{|b'\rangle\}$, determine if this property is representation-independent. Verify the assertion using spin operators and canonical observables.

The property of having strictly real matrix elements is not representation-independent. The "reality" of an operator, under the provided definition, is a basis-dependent characteristic. While the physical predictions (eigenvalues and probabilities) are invariant under unitary transformations, the specific values of the matrix elements—including their character as real or complex numbers—are not.

Consider an operator A that is "real" in the basis $\{|b'\rangle\}$, such that $A_{ij} = \langle b_i|A|b_j \rangle \in \mathbb{R}$. A change of basis to a new orthonormal set $\{|c'\rangle\}$ is governed by a unitary transformation U :

$$|c_n\rangle = U|b_n\rangle, \quad \langle c_n| = U^\dagger \langle b_n|,$$

The matrix elements in the new basis are:

$$A'_{nm} = \langle c_n|A|c_m \rangle = \sum_{ij} \langle c_n|b_i \rangle \langle b_i|A|b_j \rangle \langle b_j|c_m \rangle,$$

In matrix notation: $A' = U^\dagger A U$.

Even if A is a real symmetric matrix, if U is a complex unitary matrix (which is the general case in quantum mechanics), the product $U^\dagger A U$ will typically yield complex matrix elements.

Let us examine the operator S_y using the results from the previous discussion:

In the S_z basis $\{|+\rangle, |-\rangle\}$, the matrix representation is:

$$S_y \doteq \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

Under the proposed definition, S_y is not real in this basis because the elements are purely imaginary.



In its own eigenbasis $\{|y, +\rangle, |y, -\rangle\}$, the matrix is diagonalized:

$$S_y \doteq \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

In this basis, S_y is real. Since the "reality" changes with the basis, the concept is not representation-independent.

The Momentum Operator $\hat{p}$: In the momentum representation $\{|p\rangle\}$, the operator is diagonal and its elements are real: $\langle p|\hat{p}|p'\rangle = p\delta(p - p')$.

However, in the position representation $\{|x\rangle\}$ the matrix elements involve the kernel of the derivative:

$$\langle x|\hat{p}|x'\rangle = -i\hbar \frac{\partial}{\partial x} \delta(x - x'),$$

The explicit presence of the imaginary unit i (required to ensure $\hat{p}$ is Hermitian) demonstrates that $\hat{p}$ is not "real" in the position basis.

The reality of matrix elements is a representation-dependent property. In quantum mechanics, we prioritize Hermiticity ($A = A^\dagger$) as the relevant basis-independent property for observables, as it guarantees real eigenvalues regardless of the chosen representation.

1.28 Construct the transformation matrix that connects the S_z diagonal basis to the S_x diagonal basis. Show that your result is consistent with the general relation:

$$U = \sum_r |b^{(r)}\rangle \langle a^{(r)}|.$$

Solution:

For the transformation from S_z to S_x , we define:

$$U = |+_x\rangle \langle +_z| + |-_x\rangle \langle -_z|,$$

To prove this operator correctly "connects" the bases, we analyze its action on the initial state space. Using the orthonormality of the S_z basis ($\langle i_z | j_z \rangle = \delta_{ij}$):

$$U|+_z\rangle = |+_x\rangle \langle +_z|+_z\rangle + |-_x\rangle \langle -_z|+_z\rangle = |+_x\rangle,$$

$$U|-_z\rangle = |+_x\rangle \langle +_z|-_z\rangle + |-_x\rangle \langle -_z|-_z\rangle = |-_x\rangle,$$

The operator U is, by definition, the map that satisfies $|b^{(r)}\rangle = U|a^{(r)}\rangle$.



The transformation matrix U that connects basis A to basis B is defined such that its columns are the components of the new basis vectors ($|\pm_x\rangle$) expressed in the old basis ($|\pm_z\rangle$). Thus:

$$U \doteq \begin{pmatrix} \langle +_z | +_x \rangle & \langle +_z | -_x \rangle \\ \langle -_z | +_x \rangle & \langle -_z | -_x \rangle \end{pmatrix},$$

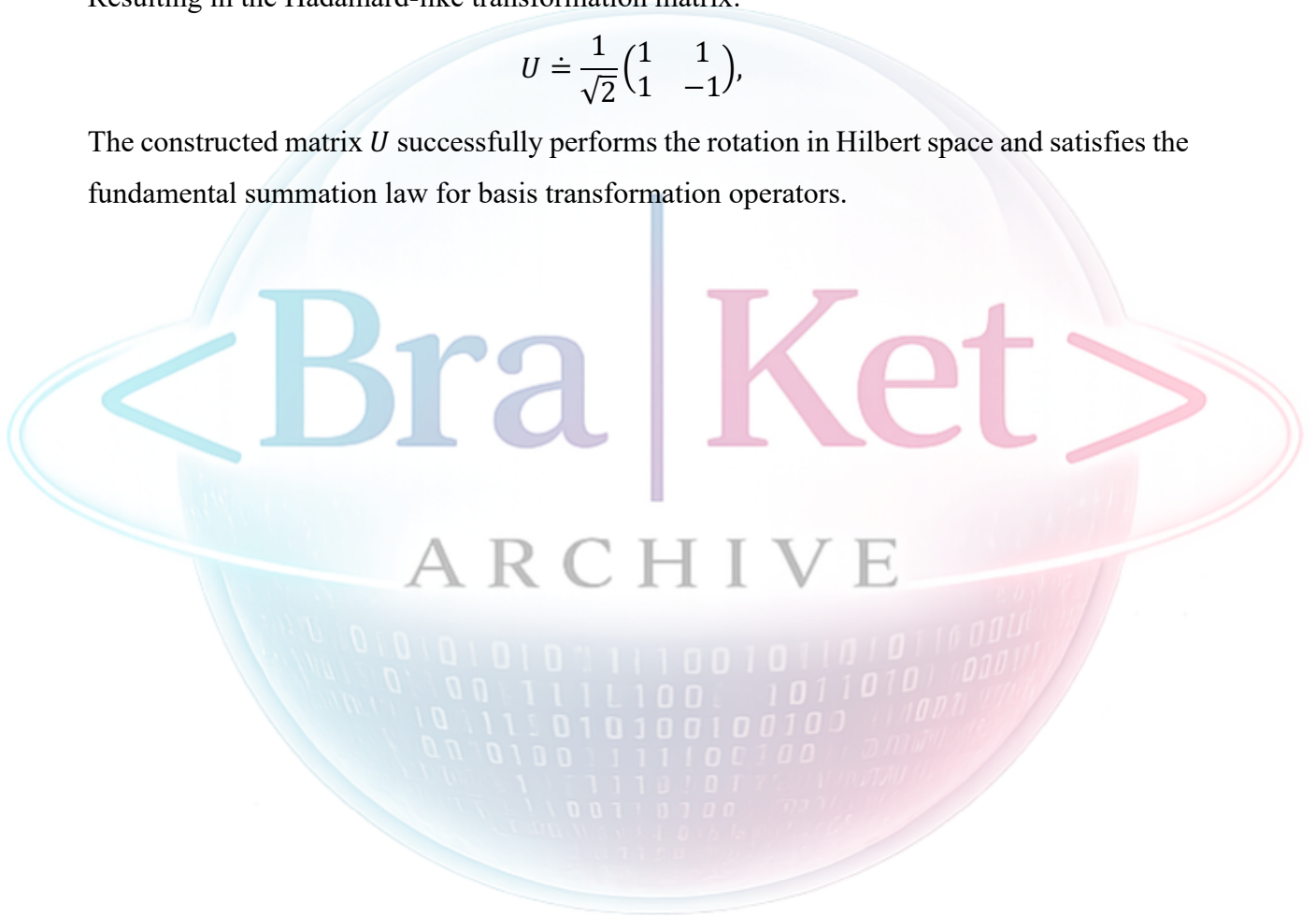
Using $|\pm_x\rangle = \frac{1}{\sqrt{2}}[|+_z\rangle \pm |-_z\rangle]$:

$$U \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} \langle +_z | (|+_z\rangle + |-_z\rangle) & \langle +_z | (|+_z\rangle - |-_z\rangle) \\ \langle -_z | (|+_z\rangle + |-_z\rangle) & \langle -_z | (|+_z\rangle - |-_z\rangle) \end{pmatrix},$$

Resulting in the Hadamard-like transformation matrix:

$$U \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

The constructed matrix U successfully performs the rotation in Hilbert space and satisfies the fundamental summation law for basis transformation operators.





1.29 a. Suppose that $f(A)$ is a function of a Hermitian operator A with the property $A|a'\rangle = a'|a'\rangle$. Evaluate $\langle b''|f(A)|b'\rangle$ when the transformation matrix from the a' basis to the b' basis is known.

b. Using the continuum analogue of the result obtained in (a), evaluate:

$$\langle \vec{p}''|F(r)|\vec{p}'\rangle.$$

Simplify your expression as far as you can. Note that r is $\sqrt{x^2 + y^2 + z^2}$, where x , y , and z are operators.

Solution:

a) Matrix Elements of Operator Functions

Let A be a Hermitian operator with a complete set of orthonormal eigenkets $\{|a'\rangle\}$ and corresponding eigenvalues $\{a'\}$. Any function of the operator $f(A)$ is defined through its spectral decomposition as:

$$f(A) = \sum_{a'} f(a') |a'\rangle \langle a'|,$$

To evaluate the matrix element in the b -representation, we insert the resolution of identity (completeness relation) $\mathbb{I} = \sum_{a'} |a'\rangle \langle a'|$ twice, or more directly, use the definition of $f(A)$ as follows:

$$\langle b''|f(A)|b'\rangle = \langle b''| \left(\sum_{a'} f(a') |a'\rangle \langle a'| \right) |b'\rangle,$$

By the linearity of the inner product:

$$\langle b''|f(A)|b'\rangle = \sum_{a'} f(a') \langle b''|a'\rangle \langle a'|b'\rangle,$$

Let $U_{b''a'} = \langle b''|a'\rangle$ be the elements of the unitary transformation matrix from the a -basis to the b -basis. The expression becomes:

$$\langle b''|f(A)|b'\rangle = \sum_{a'} f(a') U_{b''a'} U_{b'a'}^*,$$

The matrix element is a weighted sum of the function values $f(a')$, where the weights are determined by the overlaps (transformation matrix elements) between the initial and final basis states.

b) Continuum Analogue for Position and Momentum



In the continuum limit, the resolution of identity in the position basis $\{|x\rangle\}$ is given by:

$$\mathbb{I} = \int |x\rangle\langle x| d^3x,$$

Since $F(r)$ is a function of the position operator $\hat{x}$, it is diagonal in the position representation: $F(r)|x\rangle = F(|x|)|x\rangle$. We insert the identity operator:

$$\langle \vec{p}'' | F(r) | \vec{p}' \rangle = \int \langle \vec{p}'' | x \rangle \langle x | F(r) | \vec{p}' \rangle d^3x = \int F(|x|) \langle \vec{p}'' | x \rangle \langle x | \vec{p}' \rangle d^3x,$$

Using the standard plane-wave expansion for the transformation between position and momentum bases:

$$\langle x | \vec{p} \rangle = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{i\frac{\vec{p}\cdot\vec{x}}{\hbar}},$$

Substituting this into the integral:

$$\begin{aligned} \langle \vec{p}'' | F(r) | \vec{p}' \rangle &= \int F(x) \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{-i\frac{\vec{p}''\cdot\vec{x}}{\hbar}} \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{i\frac{\vec{p}'\cdot\vec{x}}{\hbar}} d^3x, \\ \langle \vec{p}'' | F(r) | \vec{p}' \rangle &= \frac{1}{(2\pi\hbar)^3} \int F(x) \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} e^{-i\frac{(\vec{p}''-\vec{p}')\cdot\vec{x}}{\hbar}} d^3x, \end{aligned}$$

Let $\vec{q} = \frac{\vec{p}''-\vec{p}'}{\hbar}$ be the wave-vector transfer. The integral is essentially a 3D Fourier Transform of the spatial function $F(r)$. For a purely radial function $F(r)$, we can use spherical coordinates (r, θ, ϕ) where the z-axis of the integral is aligned with $\vec{q}$:

$$\langle \vec{p}'' | F(r) | \vec{p}' \rangle = \frac{1}{(2\pi\hbar)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} F(r) r^2 \sin(\theta) e^{-iqr \cos(\theta)} dr d\theta d\phi,$$

Let $U = \cos(\theta)$. Then $du = -\sin(\theta) d\theta$.

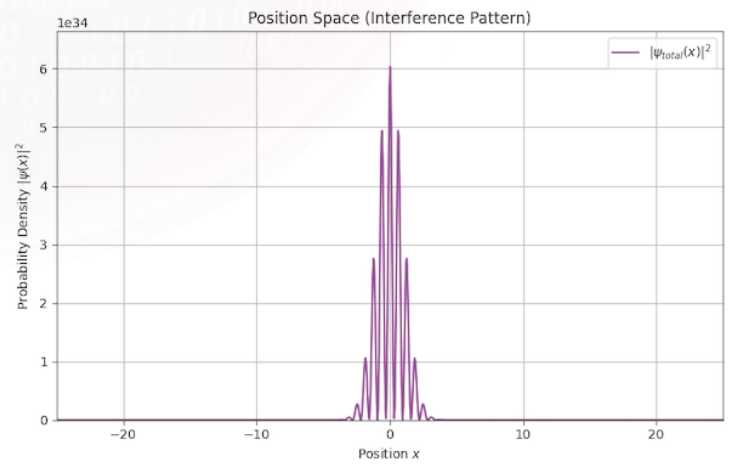
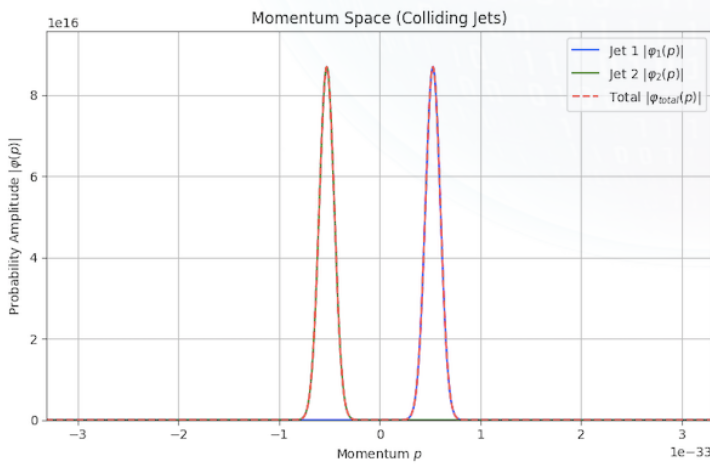
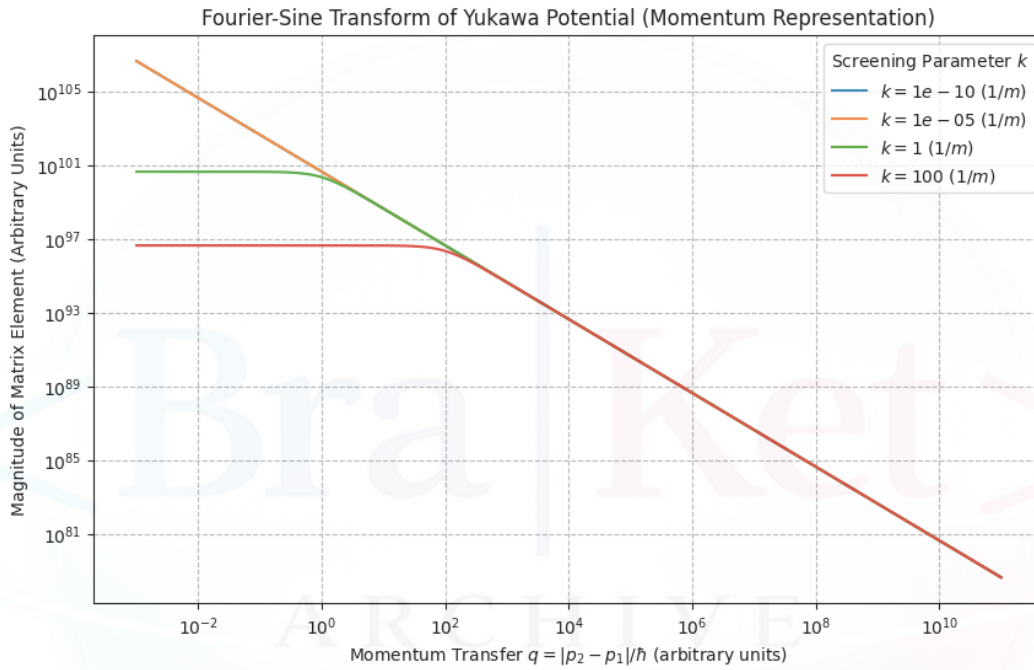
The limits of integration change from $[0, \pi]$ to $[\cos(0), \cos(\pi)] = [1, -1]$.

$$\begin{aligned} \langle \vec{p}'' | F(r) | \vec{p}' \rangle &= \frac{2\pi}{(2\pi\hbar)^3} \int_{-1}^1 \int_0^\infty F(r) r^2 e^{-iqr u} dr du, \\ \langle \vec{p}'' | F(r) | \vec{p}' \rangle &= \frac{2\pi}{(2\pi\hbar)^3} \int_0^\infty F(r) \frac{r^2}{-iqr} (e^{-iqr} - e^{iqr}) dr, \\ \langle \vec{p}'' | F(r) | \vec{p}' \rangle &= \frac{1}{2\pi^2 \hbar^3 q} \int_0^\infty r F(r) \sin(qr) dr, \end{aligned}$$



where $q = \frac{|\vec{p}'' - \vec{p}'|}{\hbar}$.

The matrix element $\langle \vec{p}'' | F(r) | \vec{p}' \rangle$ is proportional to the 3D Fourier Transform of the spatial potential $F(r)$. For spherically symmetric distributions, this reduces to a Fourier-Sine transform which depends exclusively on the momentum transfer $\vec{q} = \frac{\vec{p}'' - \vec{p}'}{\hbar}$. This result establishes that scattering amplitudes in the momentum representation are governed by the spectral decomposition of the interaction potential.





1.30 a. Let x and p_x be the coordinate and linear momentum in one dimension. Evaluate the classical Poisson bracket:

$$[x, F(p_x)]_{classical}.$$

b. Let x and p_x be the corresponding quantum-mechanical operators this time. Evaluate the commutator:

$$\left[x, \exp\left(\frac{ip_x a}{\hbar}\right) \right].$$

c. Using the result obtained in (b), prove that:

$$\exp\left(\frac{ip_x a}{\hbar}\right) |x'\rangle, \quad (x|x'\rangle = x'|x'\rangle),$$

is an eigenstate of the coordinate operator x . What is the corresponding eigen-value?

Solution:

a) Classical Poisson Bracket

In Hamiltonian mechanics, the Poisson bracket of two functions f and g in phase space is defined as:

$$[f, g]_{p,q} = \frac{\partial f}{\partial x} \frac{\partial g}{\partial p_x} - \frac{\partial f}{\partial p_x} \frac{\partial g}{\partial x},$$

Setting $f = x$ and $g = F(p_x)$:

$$\begin{aligned} [x, F(p_x)] &= \frac{\partial x}{\partial x} \frac{\partial F(p_x)}{\partial p_x} - \frac{\partial x}{\partial p_x} \frac{\partial F(p_x)}{\partial x}, \\ \therefore [x, F(p_x)] &= \frac{dF(p_x)}{dp_x} = F'(p_x). \end{aligned}$$

b) Quantum Mechanical Commutator

Let $T(a) = \exp\left(\frac{ip_x a}{\hbar}\right)$ be the translation operator. To evaluate the commutator, we use the fundamental relation $[\hat{x}, \hat{p}_x] = i\hbar$. A useful identity for an operator A and a function of an operator $g(B)$ is:

$$[A, g(B)] = [A, B] \frac{dg}{dB},$$

In this case:

$$\begin{aligned} [\hat{x}, T(a)] &= [\hat{x}, \hat{p}_x] \frac{d}{d\hat{p}_x} \left[\exp\left(\frac{ip_x a}{\hbar}\right) \right] = i\hbar \frac{ia}{\hbar} \left[\exp\left(\frac{ip_x a}{\hbar}\right) \right] = -a \exp\left(\frac{ip_x a}{\hbar}\right), \\ \therefore [\hat{x}, T(a)] &= -aT(a), \end{aligned}$$



c) Eigenstates of the Coordinate Operator

We wish to determine the action of $\hat{x}$ on the state $|\Psi\rangle = T(a)|x'\rangle$. Using the commutator result from part (b):

$$\hat{x}T(a) - T(a)\hat{x} = -aT(a) \Rightarrow \hat{x}T(a) = T(a)\hat{x} - aT(a),$$

Applying this to the position eigenket $|x'\rangle$:

$$\hat{x}(T(a)|x'\rangle) = T(a)(\hat{x}|x'\rangle) - aT(a)|x'\rangle,$$

$$\hat{x}(T(a)|x'\rangle) = (x' - a)T(a)|x'\rangle,$$

The state $\exp\left(\frac{ip_x a}{\hbar}\right)|x'\rangle$ is indeed an eigenstate of the coordinate operator $\hat{x}$ with the corresponding eigenvalue $(x' - a)$. Physical Insight: The operator $\exp\left(\frac{ip_x a}{\hbar}\right)$ shifts the position of the state by a distance $-a$. Note that the sign in the exponential $(+i)$ leads to a displacement in the negative direction, consistent with the active transformation convention.

1.31 a. On p.247, Gottfried (1966) states that:

$$[x_i, G(\vec{p})] = i\hbar \frac{\partial G}{\partial p_i}, \quad [p_i, F(\vec{x})] = -i\hbar \frac{\partial F}{\partial x_i},$$

can be “easily derived” from the fundamental commutation relations for all functions of F and G that can be expressed as power series in their arguments. Verify this statement.

b. Evaluate $[x^2, p^2]$. Compare your result with the classical Poisson bracket $[x^2, p^2]_{\text{classical}}$.

Solution:

a) Rigorous Proof for Power Series Representations

Let $\hat{x}_i$ and $\hat{p}_j$ satisfy the canonical commutation relations $[x_i, p_j] = i\hbar\delta_{ij}$. For any functions $F(\vec{x})$ and $G(\vec{p})$ analytic in their arguments:

$$[x_i, G(\vec{p})] = i\hbar \frac{\partial G}{\partial p_i}, \quad [p_i, F(\vec{x})] = -i\hbar \frac{\partial F}{\partial x_i},$$

Proof via Power Series Expansion: Lemma: Show that $[x_i, p_j^n] = i\hbar\delta_{ij}np_j^{n-1}$ using mathematical induction.

Base case ($n = 1$): $[x_i, p_j] = i\hbar\delta_{ij}$, which holds by definition.

Inductive step: Assume the identity holds for n . Consider $n + 1$:

$$[x_i, p_j^{n+1}] = [x_i, p_j^n p_j] = p_j^n [x_i, p_j] + p_j [x_i, p_j^n] = p_j^n i\hbar\delta_{ij} + i\hbar\delta_{ij} n p_j^n,$$



$$\therefore [x_i, p_j^{n+1}] = i\hbar \delta_{ij}(n+1)p_j^n,$$

The lemma is proved. Thus, $[x_i, p_i^n] = i\hbar \frac{\partial}{\partial p_i}(p_i^n)$.

Generalization to $G(\vec{p})$:

Since $G(\vec{p})$ is analytic, expand it as a multi-variable power series:

$$G(\vec{p}) = \sum_n c_n p_1^{n_1} p_2^{n_2} p_3^{n_3},$$

By the linearity of the commutator and noting that $[x_i, p_j] = 0$ for $i \neq j$:

$$[x_i, G(\vec{p})] = \sum_n c_n p_1^{n_1} \dots [x_i, p_i^{n_i}] \dots p_3^{n_3},$$

$$[x_i, G(\vec{p})] = i\hbar \sum_n c_n p_1^{n_1} \dots (n_i p_i^{n_i-1}) \dots p_3^{n_3} = i\hbar \frac{\partial G}{\partial p_i},$$

Following an identical series expansion for $F(\vec{x})$ and noting $[p_i, x_j] = -i\hbar \delta_{ij}$:

$$\therefore [p_i, F(\vec{x})] = -i\hbar \frac{\partial F}{\partial x_i}.$$

b) Evaluation and Classical Comparison of $[x^2, p^2]$

Using the algebraic identity $[A, BC] = [A, B]C + B[A, C]$:

$$[x^2, p^2] = [x^2, p]p + p[x^2, p],$$

Employing the theorem proved in Part (a) where $[x^2, p] = -[p, x^2] = i\hbar \frac{\partial}{\partial x} x^2 = 2i\hbar x$:

$$[x^2, p^2] = [2i\hbar x]p + p[2i\hbar x] = 2i\hbar(xp + px),$$

To express this in standard ordered form, substitute $px = xp - i\hbar$:

$$\therefore [x^2, p^2] = 4i\hbar xp + 2\hbar^2.$$

In classical phase space, the Poisson bracket is defined as:

$$[x^2, p^2]_{\text{classical}} = \frac{\partial x^2}{\partial x} \frac{\partial p^2}{\partial p} - \frac{\partial x^2}{\partial p} \frac{\partial p^2}{\partial x} = 4xp,$$

According to Dirac's canonical quantization rule, the quantum commutator map relates to the classical Poisson bracket via:

$$[A, B] = i\hbar [A, B]_{\text{classical}} + O(\hbar^2),$$

Multiplying the classical result by $i\hbar$ yields:

$$[x^2, p^2] = 4i\hbar xp,$$



Comparing the two results:

$$[x^2, p^2] = i\hbar[x^2, p^2]_{\text{classical}} + 2\hbar^2,$$

The quantum commutator contains a purely quantum correction term ($2\hbar^2$). This higher-order correction arises because x and p do not commute, requiring symmetric operator ordering when transitioning from classical to quantum observables.

1.32 The translation operator for a finite (spatial) displacement is given by:

$$j(\vec{l}) = \exp\left(\frac{-i\vec{p} \cdot \vec{l}}{\hbar}\right),$$

where $\vec{p}$ is the momentum operator.

a. Evaluate:

$$[x_i, j(\vec{l})].$$

b. Using (a) (or otherwise), demonstrate how the expectation value $\langle \vec{x} \rangle$ changes under translation.

Solution:

a) Evaluation of the Commutator $[x_i, j(\vec{l})]$

Let $G(\vec{p}) = j(\vec{l}) = \exp\left(\frac{-i\vec{p} \cdot \vec{l}}{\hbar}\right)$ be an analytic function of the momentum operator p . We invoke the previously proven theorem for analytic operator functions:

$$[x_i, G(\vec{p})] = i\hbar \frac{\partial G}{\partial p_i},$$

We express the vector dot product explicitly in component form using Einstein summation notation: $\vec{p} \cdot \vec{l} = p_j l_j$. We compute the partial derivative with respect to p_i :

$$\frac{\partial}{\partial p_i} \left[\exp\left(\frac{-ip_j l_j}{\hbar}\right) \right] = \left(-\frac{il_j}{\hbar} \frac{\partial p_j}{\partial p_i} \right) \exp\left(\frac{-ip_j l_j}{\hbar}\right),$$

Using $\frac{\partial p_j}{\partial p_i} = \delta_{ji}$, the Kronecker delta collapses the sum:

$$\frac{\partial j(\vec{l})}{\partial p_i} = -\frac{il_i}{\hbar} j(\vec{l}),$$

Substituting this back into the generalized commutator relation:

$$[x_i, j(\vec{l})] = i\hbar \left(-\frac{il_i}{\hbar} \right) j(\vec{l}) = l_i j(\vec{l}),$$



$$\therefore [x_i, j(\vec{l})] = l_i j(\vec{l}) = l_i \exp\left(\frac{-i\vec{p} \cdot \vec{l}}{\hbar}\right).$$

b) Transformation of the Expectation Value $\langle \vec{x} \rangle$

Let $|\Psi\rangle$ be the state vector prior to translation. The translated state $|\Psi'\rangle$ is defined via the active transformation:

$$|\Psi'\rangle = j(\vec{l})|\Psi\rangle,$$

The expectation value of the coordinate operator component x_i under the translated state is:

$$\langle x_i \rangle' = \langle \Psi' | x_i | \Psi' \rangle = \langle \Psi | j^\dagger(\vec{l}) x_i j(\vec{l}) | \Psi \rangle,$$

We rewrite the commutator from Part (a) as an operator identity:

$$x_i j(\vec{l}) - j(\vec{l}) x_i = l_i j(\vec{l}) \Rightarrow x_i j(\vec{l}) = j(\vec{l}) x_i + l_i j(\vec{l}),$$

Since $j(\vec{l})$ is a unitary operator, its adjoint equals its inverse: $j^\dagger(\vec{l}) = j^{-1}(\vec{l})$. We multiply the identity by $j^\dagger(\vec{l})$ from the left:

$$j^\dagger(\vec{l}) x_i j(\vec{l}) = j^\dagger(\vec{l}) j(\vec{l}) x_i + l_i j^\dagger(\vec{l}) j(\vec{l}),$$

$$j^\dagger(\vec{l}) x_i j(\vec{l}) = x_i + l_i \mathbb{I},$$

We substitute this similarity transformation back into the expectation value integral:

$$\langle x_i \rangle' = \langle \Psi | x_i + l_i \mathbb{I} | \Psi \rangle = \langle \Psi | x_i | \Psi \rangle + l_i \langle \Psi | \Psi \rangle,$$

Assuming normalized states ($\langle \Psi | \Psi \rangle = 1$):

$$\langle x_i \rangle' = \langle x_i \rangle + l_i,$$

Generalizing to full vector notation, the spatial expectation value shifts linearly by the displacement vector:

$$\langle \vec{x} \rangle \xrightarrow{j(\vec{l})} \langle \vec{x} \rangle + \vec{l},$$

The relation $\langle \vec{x} \rangle \xrightarrow{j(\vec{l})} \langle \vec{x} \rangle + \vec{l}$ establishes that the unitary operator $j(\vec{l})$ structurally shifts the physical state across space while preserving the geometric properties of the Hilbert space.

The non-zero commutator $[x_i, j(\vec{l})] = l_i j(\vec{l})$ proves that $\vec{p}$ acts as the infinitesimal generator of translations, demonstrating that spatial homogeneity in quantum mechanics is governed by the algebraic structure of the canonical commutation relations.



1.33 In the main text we discussed the effect of $j(d\vec{x}')$ on the position and momentum eigenkets and on a more general state ket $|\alpha\rangle$. We can also study the behavior of expectation values $\langle\vec{x}\rangle$ and $\langle\vec{p}\rangle$ under infinitesimal translation. Using $[\vec{x}, j(d\vec{x}')] = d\vec{x}'$, $[\vec{p}, j(d\vec{x}')] = 0$ and $|\alpha\rangle \rightarrow j(d\vec{x})|\alpha\rangle$ only, prove $\langle\vec{x}\rangle \rightarrow \langle\vec{x}\rangle + d\vec{x}'$, $\langle\vec{p}\rangle \rightarrow \langle\vec{p}\rangle$ under infinitesimal translation.

Solution :

The transformed expectation value under the shifted state $|\alpha'\rangle = j(d\vec{x}')|\alpha\rangle$ is given by:

$$\langle\vec{x}\rangle' = \langle\alpha'|\vec{x}|\alpha'\rangle = \langle\alpha|j^\dagger(d\vec{x}')\vec{x}j(d\vec{x}')|\alpha\rangle,$$

Using the commutator we have $\vec{x}j(d\vec{x}') = j(d\vec{x}')\vec{x} + d\vec{x}'$, we rewrite the operator sandwich:

$$j^\dagger(d\vec{x}')\vec{x}j(d\vec{x}') = j^\dagger(d\vec{x}')j(d\vec{x}')\vec{x} + j^\dagger(d\vec{x}')d\vec{x}',$$

Since $j(d\vec{x}') = \mathbb{I} - i\vec{p} \cdot \frac{d\vec{x}'}{\hbar}$:

$$\begin{aligned} j^\dagger(d\vec{x}')\vec{x}j(d\vec{x}') &= \left(\mathbb{I} + \left|\vec{p} \cdot \frac{d\vec{x}'}{\hbar}\right|^2\right)\vec{x} + \left(\mathbb{I} + i\vec{p} \cdot \frac{d\vec{x}'}{\hbar}\right)d\vec{x}', \\ j^\dagger(d\vec{x}')\vec{x}j(d\vec{x}') &= \vec{x} + \mathbb{I}d\vec{x}' + \left(\left|\vec{p} \cdot \frac{d\vec{x}'}{\hbar}\right|^2\right)\vec{x} + \left(i\vec{p} \cdot \frac{d\vec{x}'}{\hbar}\right)d\vec{x}', \end{aligned}$$

Dropping higher-order infinitesimals:

$$\langle\vec{x}\rangle' = \langle\alpha|\vec{x} + \mathbb{I}d\vec{x}'|\alpha\rangle = \langle\alpha|\vec{x}|\alpha\rangle + d\vec{x}'\langle\alpha|\alpha\rangle,$$

Assuming a normalized ket ($\langle\alpha|\alpha\rangle = 1$), vector generalization yields:

$$\langle\vec{x}\rangle \xrightarrow{j(d\vec{x}')} \langle\vec{x}\rangle + d\vec{x}',$$

Following the same active transformation layout for the momentum operator:

$$\langle\vec{p}\rangle' = \langle\alpha'|\vec{p}|\alpha'\rangle = \langle\alpha|j^\dagger(d\vec{x}')\vec{p}j(d\vec{x}')|\alpha\rangle,$$

Since $[\vec{p}, j(d\vec{x}')] = 0$, the operators commute natively $\vec{p}j(d\vec{x}') = j(d\vec{x}')\vec{p}$:

$$j^\dagger(d\vec{x}')\vec{p}j(d\vec{x}') = j^\dagger(d\vec{x}')j(d\vec{x}')\vec{p},$$

Since $j(d\vec{x}') = \mathbb{I} - i\vec{p} \cdot \frac{d\vec{x}'}{\hbar}$:

$$j^\dagger(d\vec{x}')\vec{p}j(d\vec{x}') = \left(\mathbb{I} + \left|\vec{p} \cdot \frac{d\vec{x}'}{\hbar}\right|^2\right)\vec{p},$$



Dropping higher-order infinitesimals:

$$\langle \vec{p} \rangle' = \langle \alpha | \vec{p} | \alpha \rangle = \langle \vec{p} \rangle,$$

Assuming a normalized ket ($\langle \alpha | \alpha \rangle = 1$), vector generalization yields:

$$\langle \vec{p} \rangle \xrightarrow{\delta(d\vec{x}')} \langle \vec{p} \rangle,$$

Under an infinitesimal spatial displacement, the expectation value of the coordinate vector transforms covariantly with the macroscopic coordinate change, whereas the expectation value of momentum remains strictly invariant. This asymmetric behavior reflects that spatial translation shifts the localization profile in the configuration domain without modifying the phase distribution in the frequency (momentum) domain.

1.34 Starting with a momentum operator $\vec{p}$ having eigenstates $|\vec{p}'\rangle$, define an infinitesimal boost operator $\mathcal{B}(d\vec{p}')$ that changes one momentum eigenstate into another, that is:

$$\mathcal{B}(d\vec{p}') |\vec{p}'\rangle = |\vec{p}' + d\vec{p}'\rangle.$$

Show that the form $\mathcal{B}(d\vec{p}') = 1 + i\vec{W} \cdot d\vec{p}'$, where $\vec{W}$ is Hermitian, satisfies the unitary, associative, and inverse properties that are appropriate for $\mathcal{B}(d\vec{p}')$. Use dimensional analysis to express $\vec{W}$ in terms of the position operator $\vec{x}$, and show that the result satisfies the canonical commutation relations $[x_i, p_j] = i\hbar\delta_{ij}$. Derive an expression for the matrix element $\langle \vec{p}' | \vec{x} | \alpha \rangle$ in terms of a derivative with respect to $\vec{p}'$ of $\langle \vec{p}' | \alpha \rangle$.

Solution:

Fundamental Algebraic Properties of $\mathcal{B}(d\vec{p}')$

Let $\mathcal{B}(d\vec{p}') = 1 + i\vec{W} \cdot d\vec{p}'$ be the infinitesimal boost operator acting on the momentum Hilbert space, where $\vec{W}$ is a vector of Hermitian operators ($\vec{W} = \vec{W}^\dagger$).

Unitary Property: To first order in the infinitesimal parameter $d\vec{p}'$, the operator must preserve the norm of state vectors.

$$\mathcal{B}^\dagger(d\vec{p}')\mathcal{B}(d\vec{p}') = (1 - i\vec{W} \cdot d\vec{p}')(1 + i\vec{W} \cdot d\vec{p}') = 1 + |\vec{W} \cdot d\vec{p}'|^2,$$

Appealing to first-order linearity, the quadratic terms vanish, yielding $\mathcal{B}^\dagger(d\vec{p}')\mathcal{B}(d\vec{p}') = 1$.

Thus, $\mathcal{B}(d\vec{p}')$ is unitary.

Inverse Property: The inverse operator must reverse the transformation, corresponding to a boost of $-d\vec{p}'$.



$$\mathcal{B}^{-1}(d\vec{p}') = \mathcal{B}(-d\vec{p}') = 1 - i\vec{W} \cdot d\vec{p}',$$

Verifying the identity property:

$$\mathcal{B}^{-1}(d\vec{p}')\mathcal{B}(d\vec{p}') = (1 - i\vec{W} \cdot d\vec{p}')(1 + i\vec{W} \cdot d\vec{p}') = 1 + |\vec{W} \cdot d\vec{p}'|^2 \approx 1,$$

Associative Property: Consider two consecutive independent infinitesimal boosts $d\vec{p}'_1$ and $d\vec{p}'_2$. The composition follows the Abelian group law for translations in momentum space:

$$\mathcal{B}(d\vec{p}'_1)[\mathcal{B}(d\vec{p}'_2)|\vec{p}\rangle] = \mathcal{B}(d\vec{p}'_1)|\vec{p} + d\vec{p}'_2\rangle = |\vec{p} + d\vec{p}'_2 + d\vec{p}'_1\rangle,$$

$$[\mathcal{B}(d\vec{p}'_1)\mathcal{B}(d\vec{p}'_2)]|\vec{p}\rangle = [1 + i\vec{W} \cdot (d\vec{p}'_1 + d\vec{p}'_2)]|\vec{p}\rangle = \mathcal{B}(d\vec{p}'_1 + d\vec{p}'_2),$$

$$|\vec{p}\rangle[\mathcal{B}(d\vec{p}'_1)\mathcal{B}(d\vec{p}'_2)]|\vec{p}\rangle = |\vec{p} + d\vec{p}'_2 + d\vec{p}'_1\rangle,$$

As matrix multiplication over linear operators is inherently associative, the composition identity matches the mapping structure.

Dimensional Analysis and Identification of $\vec{W}$

The argument of the exponent or the second term in an infinitesimal expansion must be strictly dimensionless.

Let $[\cdot]$ denote SI dimensions. We have $[d\vec{p}'] = \text{Mass} \cdot \text{Length} \cdot \text{Time}^{-1}$.

The term $[\vec{W} \cdot d\vec{p}'] = 1 \Rightarrow [\vec{W}] = [\text{Momentum}]^{-1} = \text{Mass}^{-1} \cdot \text{Length}^{-1} \cdot \text{Time}$.

Recall the dimensions of the position operator $\vec{x}$ and Planck's constant $\hbar$:

$$[\vec{x}] = \text{Length}, \quad [\hbar] = \text{Mass} \cdot \text{Length}^2 \cdot \text{Time}^{-1},$$

Forming the ratio $[\vec{x}/\hbar]$ gives:

$$\left[\frac{\vec{x}}{\hbar}\right] = \text{Mass}^{-1} \cdot \text{Length}^{-1} \cdot \text{Time},$$

Matching dimensions yields the structural ansatz: $\vec{W} = \pm \frac{\vec{x}}{\hbar}$. To adopt canonical conventions for the sign, we define the generator as:

$$\vec{W} = +\frac{\vec{x}}{\hbar} \Rightarrow \mathcal{B}(d\vec{p}') = 1 + \frac{i}{\hbar} \vec{x} \cdot d\vec{p}',$$

Verification of Canonical Commutation Relations

We examine the action of the commutator $[x_i, p_j]$ by assessing how a position eigenket transforms under a momentum shift. Alternatively, evaluating the infinitesimal shift on a momentum state:



$$\mathcal{B}(d\vec{p}')|\vec{p}'\rangle = \left(1 + \frac{i}{\hbar} \sum_j x_j dp'_j\right) |\vec{p}'\rangle = |\vec{p}' + d\vec{p}'\rangle,$$

Projecting this relation onto the position representation basis $\langle \vec{x} |$:

$$\langle \vec{x} | \left(1 + \frac{i}{\hbar} \vec{x} \cdot d\vec{p}'\right) |\vec{p}'\rangle = \langle \vec{x} | \vec{p}' + d\vec{p}'\rangle,$$

Using the standard free particle plane-wave expansion $\langle \vec{x} | \vec{p}'\rangle = \frac{1}{(2\pi\hbar)^{3/2}} \exp\left(\frac{i\vec{x} \cdot \vec{p}'}{\hbar}\right)$:

$$\left(1 + \frac{i}{\hbar} \vec{x} \cdot d\vec{p}'\right) \langle \vec{x} | \vec{p}'\rangle = \frac{1}{(2\pi\hbar)^{3/2}} \exp\left(\frac{i\vec{x} \cdot (\vec{p}' + d\vec{p}')}{\hbar}\right),$$

Expanding the right-hand side via Taylor series to first order in $d\vec{p}'$:

$$\exp\left(\frac{i\vec{x} \cdot \vec{p}'}{\hbar}\right) \exp\left(\frac{i\vec{x} \cdot d\vec{p}'}{\hbar}\right) \approx \exp\left(\frac{i\vec{x} \cdot \vec{p}'}{\hbar}\right) \left(1 + \frac{i}{\hbar} \vec{x} \cdot d\vec{p}'\right),$$

This structural duality confirms that $\vec{x}$ acts as the differential generator in the momentum domain, satisfying the core identity $x_i \rightarrow i\hbar \frac{\partial}{\partial p_i}$. Computing the fundamental commutator directly on an arbitrary wave function:

$$[x_i, p_j]\Psi = i\hbar \delta_{ij} \Psi$$

The ansatz is compatible with the foundational canonical commutation relations $[x_i, p_j] = i\hbar \delta_{ij}$.

Matrix Element Derivative Derivation

We wish to evaluate the expression $\langle \vec{p}' | \vec{x} | \alpha \rangle$. Consider the bra vector of the boosted state:

$$[\mathcal{B}(d\vec{p}')|\vec{p}']^\dagger = \langle \vec{p}' + d\vec{p}' | = \langle \vec{p}' | \mathcal{B}^\dagger(d\vec{p}') = \langle \vec{p}' | \left(1 - \frac{i}{\hbar} \vec{x} \cdot d\vec{p}'\right),$$

Projecting an arbitrary state ket $|\alpha\rangle$ onto this transformed bra:

$$\langle \vec{p}' + d\vec{p}' | \alpha \rangle = \left\langle \vec{p}' \left| \left(1 - \frac{i}{\hbar} \vec{x} \cdot d\vec{p}'\right) \right| \alpha \right\rangle = \langle \vec{p}' | \alpha \rangle - \frac{i}{\hbar} d\vec{p}' \cdot \langle \vec{p}' | \vec{x} | \alpha \rangle,$$

Rearranging the terms to isolate the matrix element expression:

$$-\frac{i}{\hbar} d\vec{p}' \cdot \langle \vec{p}' | \vec{x} | \alpha \rangle = \langle \vec{p}' + d\vec{p}' | \alpha \rangle - \langle \vec{p}' | \alpha \rangle,$$

Component-wise, let the displacement occur solely along the i -th coordinate direction ($d\vec{p}' = dp'_i \hat{e}_i$):

$$-\frac{i}{\hbar} \langle \vec{p}' | x_i | \alpha \rangle = \frac{\langle \vec{p}' + dp'_i \hat{e}_i | \alpha \rangle - \langle \vec{p}' | \alpha \rangle}{dp'_i},$$



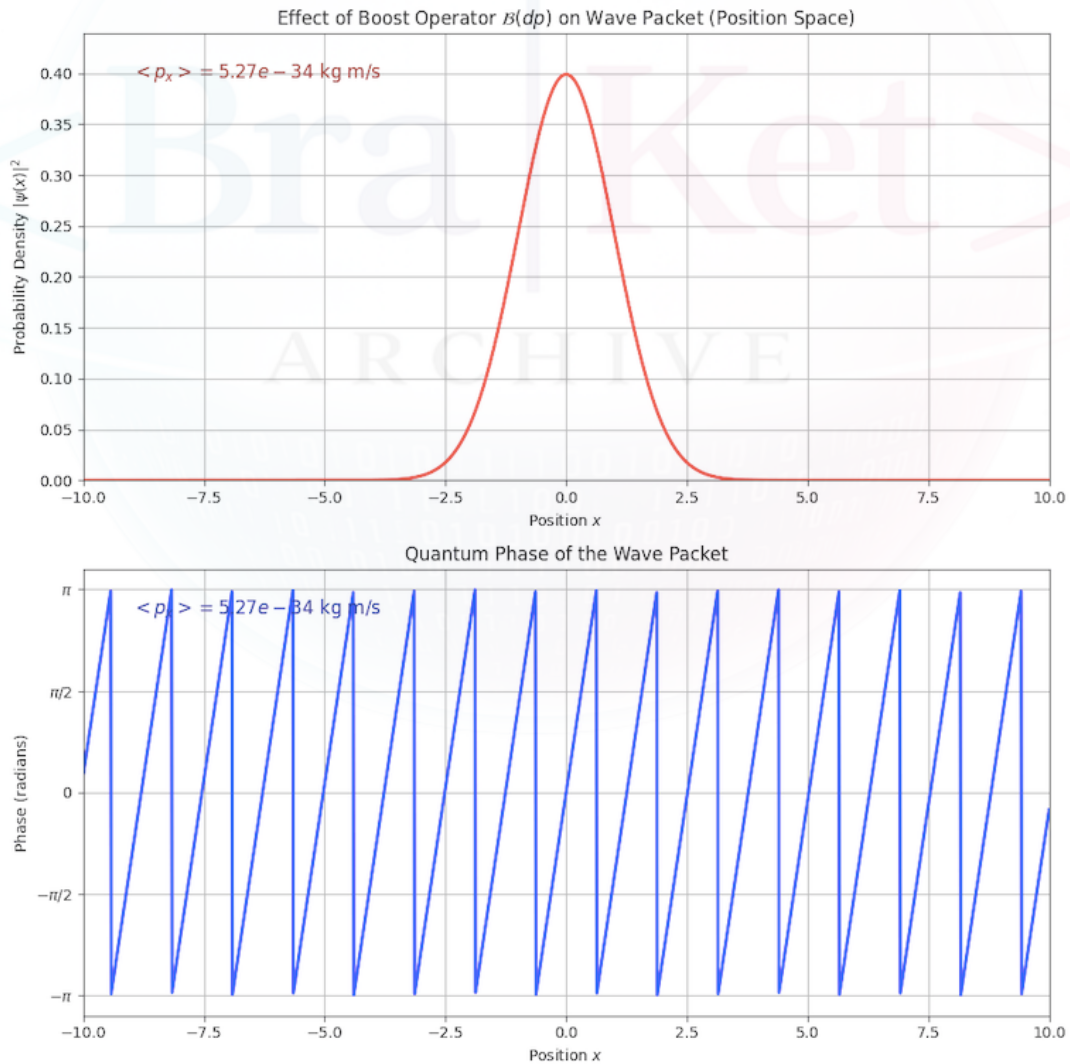
Taking the mathematical limit as $dp'_i \rightarrow 0$ recognizes the right-hand side as the formal definition of a partial derivative:

$$-\frac{i}{\hbar} \langle \vec{p}' | x_i | \alpha \rangle = \frac{\partial}{\partial p_i} \langle \vec{p}' | \alpha \rangle,$$

Multiplying by $-i\hbar$ isolates the position matrix element:

$$\langle \vec{p}' | x_i | \alpha \rangle = i\hbar \frac{\partial}{\partial p_i} \langle \vec{p}' | \alpha \rangle,$$

The position operator acting within the momentum representation behaves fundamentally as a differential operator, written as $\hat{x} \doteq i\hbar \nabla_p$. This mirrors the momentum operator's structure in the position representation $\hat{p} \doteq -i\hbar \nabla_x$, verifying the elegant, symmetric canonical mapping of Fourier conjugates in quantum mechanics.





1.35 a. Verify $\langle p \rangle = \hbar k$ and $\langle p^2 \rangle = \frac{\hbar^2}{2d^2} + \hbar^2 k^2$, for the expectation value of p and p^2 from the Gaussian wave packet :

$$\langle x' | \alpha \rangle = \left[\frac{1}{\pi^{1/4} \sqrt{d}} \right] \exp \left[ikx' - \frac{x'^2}{2d^2} \right].$$

b. Evaluate the expectation value of p and p^2 using the momentum-space wave function:

$$\langle p' | \alpha \rangle = \sqrt{\frac{d}{\hbar \sqrt{\pi}}} \exp \left[\frac{-(p' - \hbar k)^2 d^2}{2\hbar^2} \right].$$

Solution :

a) Derivation in the Position Representation

First Moment of Momentum $\langle p \rangle$

In the coordinate representation, the momentum operator is given by the differential map $p = -i\hbar \frac{\partial}{\partial x'}$. We evaluate its action on the state ket:

$$p\Psi(x') = -i\hbar \frac{\partial}{\partial x'} \left[\left[\frac{1}{\pi^{1/4} \sqrt{d}} \right] \exp \left[ikx' - \frac{x'^2}{2d^2} \right] \right] = -i\hbar \left[ik - \frac{x'}{d^2} \right] \Psi(x'),$$

$$p\Psi(x') = \left[\hbar k + i \frac{\hbar x'}{d^2} \right] \Psi(x'),$$

ARCHIVE

The expectation value is defined via the spatial inner product:

$$\langle p \rangle = \int_{-\infty}^{\infty} \Psi^*(x') p \Psi(x') dx' = \int_{-\infty}^{\infty} |\Psi(x')|^2 \left[\hbar k + i \frac{\hbar x'}{d^2} \right] dx',$$

$$\langle p \rangle = \hbar k \int_{-\infty}^{\infty} |\Psi(x')|^2 dx' + \frac{i\hbar}{d^2} \int_{-\infty}^{\infty} x' |\Psi(x')|^2 dx',$$

The spatial probability density $|\Psi(x')|^2 = \left[\frac{1}{d\sqrt{\pi}} \right] \exp \left[-\frac{x'^2}{d^2} \right]$ is strictly symmetric (even) about $x' = 0$.

The first integral normalizes to unity: $\int_{-\infty}^{\infty} |\Psi(x')|^2 dx' = 1$.

The second integrand is the product of an odd function (x') and an even function ($|\Psi(x')|^2$), causing the definite integral to vanish symmetrically.



$$\therefore \langle p \rangle = \hbar k.$$

Second Moment of Momentum $\langle p^2 \rangle$

Using the Hermiticity of p , we re-express the expectation value to avoid second derivatives:

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} (p\Psi(x'))^* (p\Psi(x')) dx' = \int_{-\infty}^{\infty} \left| \left(\hbar k + i \frac{\hbar x'}{d^2} \right) \Psi(x') \right|^2 dx',$$

Expanding the absolute square of the complex scalar factor ($|A + iB|^2 = A^2 + B^2$):

$$\begin{aligned} \langle p^2 \rangle &= \int_{-\infty}^{\infty} \left[\hbar^2 k^2 + \frac{\hbar^2 x'^2}{d^4} \right] |\Psi(x')|^2 dx', \\ \langle p^2 \rangle &= \hbar^2 k^2 \int_{-\infty}^{\infty} |\Psi(x')|^2 dx' + \frac{\hbar^2}{d^4} \int_{-\infty}^{\infty} x'^2 |\Psi(x')|^2 dx', \end{aligned}$$

We invoke the canonical Gaussian integral identity $\int_{-\infty}^{\infty} y^2 e^{-ay^2} dy$, $dy = \frac{\sqrt{\pi}}{2a^{3/2}}$ with $a = 1/d^2$:

$$\int_{-\infty}^{\infty} x'^2 |\Psi(x')|^2 dx' = \frac{1}{\sqrt{\pi}d} \left(\frac{\sqrt{\pi}d^3}{2} \right) = \frac{d^2}{2},$$

Substituting this variance back into the main expression:

$$\therefore \langle p^2 \rangle = \hbar^2 k^2 + \frac{\hbar^2}{d^4} \left(\frac{d^2}{2} \right) = \frac{\hbar^2}{2d^2} + \hbar^2 k^2,$$

b) Evaluation in the Momentum Representation

First Moment of Momentum $\langle p \rangle$

In the momentum representation, the operator p acts natively as a multiplication scalar p' .

The expectation value becomes:

$$\langle p \rangle = \int_{-\infty}^{\infty} p' |\Phi(p')|^2 dp' = \frac{d}{\hbar\sqrt{\pi}} \int_{-\infty}^{\infty} p' \exp \left[\frac{-(p' - \hbar k)^2 d^2}{\hbar^2} \right] dp'$$

We execute a linear transformation of variables to center the Gaussian profile. Let $u = p' - \hbar k \Rightarrow p' = u + \hbar k$, with $dp' = du$:

$$\langle p \rangle = \frac{d}{\hbar\sqrt{\pi}} \int_{-\infty}^{\infty} (u + \hbar k) \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du,$$



$$\langle p \rangle = \frac{d}{\hbar\sqrt{\pi}} \left[\int_{-\infty}^{\infty} u \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du + \hbar k \int_{-\infty}^{\infty} \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du \right],$$

The first integral features an asymmetric integrand and vanishes: $\int_{-\infty}^{\infty} u e^{-au^2} du = 0$.

The second integral evaluates via $\int_{-\infty}^{\infty} e^{-au^2} du = \sqrt{\pi/a}$ with $a = d^2/\hbar^2$:

$$\therefore \langle p \rangle = \frac{d}{\hbar\sqrt{\pi}} \left[0 + \hbar k \left(\frac{\hbar\sqrt{\pi}}{d} \right) \right] = \hbar k.$$

Second Moment of Momentum $\langle p^2 \rangle$

Applying the same algebraic substitution layout to the second power observable:

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} p'^2 |\phi(p')|^2 dp' = \frac{d}{\hbar\sqrt{\pi}} \int_{-\infty}^{\infty} (u + \hbar k)^2 \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du,$$

Expanding the polynomial term:

$$\langle p^2 \rangle = \frac{d}{\hbar\sqrt{\pi}} \int_{-\infty}^{\infty} (u^2 + 2u\hbar k + \hbar^2 k^2) \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du,$$

Dropping the odd cross-term ($2u\hbar k$) by symmetric integration:

$$\langle p^2 \rangle = \frac{d}{\hbar\sqrt{\pi}} \left[\int_{-\infty}^{\infty} u^2 \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du + \hbar^2 k^2 \int_{-\infty}^{\infty} \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du \right],$$

Evaluating the definite integral for u^2 with $a = d^2/\hbar^2$:

$$\int_{-\infty}^{\infty} u^2 \exp \left[\frac{-u^2 d^2}{\hbar^2} \right] du = \frac{\sqrt{\pi}}{2} \left(\frac{\hbar^2}{d^2} \right)^{\frac{3}{2}} = \frac{\sqrt{\pi} \hbar^3}{2d^3},$$

Assembling the terms with the normalization factor:

$$\langle p^2 \rangle = \frac{d}{\hbar\sqrt{\pi}} \left[\frac{\sqrt{\pi} \hbar^3}{2d^3} + \hbar^2 k^2 \left(\frac{\hbar\sqrt{\pi}}{d} \right) \right] = \frac{\hbar^2}{2d^2} + \hbar^2 k^2 \left(\frac{\hbar\sqrt{\pi}}{d} \right).$$

Both representations yield identical results, proving the unitary invariance of expectation values under the Fourier transform mapping between coordinate and momentum spaces. This consistency verifies the structural closure of the Gaussian state as a minimum uncertainty wave packet.



1.36 a. Prove the following:

$$(i) \langle p'|x|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \langle p'|\alpha \rangle,$$

$$(ii) \langle \beta|x|\alpha \rangle = \int dp' \phi_{\beta}^*(p') i\hbar \frac{\partial}{\partial p'} \phi_{\alpha}(p'),$$

where $\phi_{\alpha}(p') = \langle p'|\alpha \rangle$ and $\phi_{\beta}(p') = \langle p'|\beta \rangle$ are momentum-space wave functions.

b. What is the physical significance of :

$$\exp\left(\frac{ix\Xi}{\hbar}\right),$$

where x is the position operator and Ξ is some number with the dimension of momentum? Justify your answer.

Solution:

a) Formal Proofs of Matrix Elements

$$(i) \langle p'|x|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \langle p'|\alpha \rangle:$$

Methodology via Completeness Relation:

To evaluate the matrix element of the position operator x in the momentum continuous basis, we insert the resolution of identity in the position representation, $\mathbb{I} = \int_{-\infty}^{\infty} |x'\rangle \langle x'| dx'$:

$$\langle p'|x|\alpha \rangle = \langle p'|x \left(\int_{-\infty}^{\infty} |x'\rangle \langle x'| dx' \right) |\alpha \rangle,$$

By the linearity of the inner product and noting that $x|x'\rangle = x'|x'\rangle$:

$$\langle p'|x|\alpha \rangle = \int_{-\infty}^{\infty} x' \langle p'|x'\rangle \langle x'|\alpha \rangle dx',$$

We invoke the standard transformation kernel (transformation matrix element) between the position and momentum representations, which corresponds to the plane-wave function:

$$\langle x'|p' \rangle = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{ip'x'}{\hbar}\right) \Rightarrow \langle p'|x' \rangle = \langle x'|p' \rangle^* = \frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{ip'x'}{\hbar}\right),$$

Observe that the derivative of the kernel with respect to p' generates the x' coordinate factor:

$$\frac{\partial}{\partial p'} \langle p'|x' \rangle = \frac{\partial}{\partial p'} \left[\frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{ip'x'}{\hbar}\right) \right] = \left(-\frac{ix'}{\hbar}\right) \langle p'|x' \rangle,$$



$$x' \langle p'|x' \rangle = i\hbar \frac{\partial}{\partial p'} \langle p'|x' \rangle,$$

Substituting this differential identity back into the integral:

$$\langle p'|x|\alpha \rangle = \int_{-\infty}^{\infty} \left(i\hbar \frac{\partial}{\partial p'} \langle p'|x' \rangle \right) \langle x'|\alpha \rangle dx',$$

Since the differential operator $\frac{\partial}{\partial p'}$ acts exclusively on the variable p' , it can be factored completely outside the spatial integration domain:

$$\langle p'|x|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \int_{-\infty}^{\infty} \langle p'|x' \rangle \langle x'|\alpha \rangle dx',$$

Collapsing the identity operator gives the desired momentum-space state function:

$$\therefore \langle p'|x|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \langle p'|\alpha \rangle.$$

$$(ii) \langle \beta|x|\alpha \rangle = \int dp' \phi_{\beta}^*(p') i\hbar \frac{\partial}{\partial p'} \phi_{\alpha}(p'):$$

We expand the full transition matrix element $\langle \beta|x|\alpha \rangle$ by inserting the completeness relation for the momentum basis, $\mathbb{I} = \int_{-\infty}^{\infty} |p'\rangle \langle p'| dp'$:

$$\langle \beta|x|\alpha \rangle = \int_{-\infty}^{\infty} \langle \beta|p' \rangle \langle p'|x|\alpha \rangle dp',$$

We substitute the definition of the momentum wave function $\phi_{\beta}^*(p') = \langle \beta|p' \rangle = \langle p'|\beta \rangle^*$ and invoke the identity proven in part (i) for the remaining term:

$$\langle p'|x|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \langle p'|\alpha \rangle = i\hbar \frac{\partial}{\partial p'} \phi_{\alpha}^*(p'),$$

Assembling the terms inside the integrand:

$$\therefore \langle \beta|x|\alpha \rangle = \int_{-\infty}^{\infty} \phi_{\beta}^*(p') \left(i\hbar \frac{\partial}{\partial p'} \phi_{\alpha}^*(p') \right) dp'.$$

b) Physical Significance of $\exp\left(\frac{ix\Xi}{\hbar}\right)$

The unitary operator $\mathcal{B}(\Xi) \equiv \exp\left(\frac{ix\Xi}{\hbar}\right)$ represents a Galilean Boost Operator (or momentum translation operator) that imparts a discrete, uniform momentum shift Ξ to any state vector in the Hilbert space.



To formally justify its action, we examine how it transforms a momentum state wave function. Let $|\alpha'\rangle = \mathcal{B}(\Xi)|\alpha\rangle$. The new wave function in momentum space is:

$$\phi_\alpha(p') = \langle p' | \mathcal{B}(\Xi) | \alpha \rangle = \left\langle p' \left| \exp\left(\frac{ix\Xi}{\hbar}\right) \right| \alpha \right\rangle,$$

Expanding the exponential operator via Taylor series and using the derivative property from part (a-i) where $x \doteq i\hbar \frac{\partial}{\partial p'}$:

$$\phi_{\alpha'}(p') = \exp\left(\frac{i}{\hbar}\left(i\hbar \frac{\partial}{\partial p'}\right)\Xi\right)\phi_\alpha(p') = \exp\left(-\Xi \frac{\partial}{\partial p'}\right)\phi_\alpha(p')$$

The operator $\exp\left(-\Xi \frac{\partial}{\partial p'}\right)$ is the formal definition of the spatial shift generator (Taylor shift operator) acting on the continuous momentum variable p' . Therefore, it shifts the argument of the wave function by exactly $+\Xi$:

$$\phi_\alpha(p') = \phi_\alpha(p' - \Xi),$$

The operator actively translates the entire momentum spectrum of the system by a constant value Ξ ($\langle p \rangle = \langle p \rangle + \Xi$), while leaving the local spatial probability density profile $|\Psi(x')|^2$ unaltered. This corresponds physically to shifting the system into a moving frame of reference with a boost momentum Ξ .

ARCHIVE